复旦数学资料汇编
几何测度论

来源：香蕉空间 Solution 资料库

2026 年 10 月 13 日

编者说明

本册按课程汇集原站讲义、笔记、习题及试卷，保持原资料内容。格式转换不构成数学证明审核。每篇保留来源网址；原站未建立的页面不编造正文。内容采用原站标注的 CC BY-SA 4.0 许可，署名与详细历史可由原网页查看。

目录

第一章 几何测度论 **1**

1.1 习题: 几何测度论 . . . 1
1.2 Dimensions . . . 1
1.3 Orthogonal projections . . . 2
1.4 Differentiation of measures . . . 2
1.5 Fourier Transforms . . . 3
1.6 Self–similar sets . . . 4
1.7 讲义: 几何测度论 . . . 4
1.8 讲义: 几何测度论/ Chapter 1 Measure Theory . . . 5
1.9 Middle-α Cantor Sets . . . 5
1.10 Minkovski Dimension . . . 6
1.11 Measures . . . 7
1.11.1 Radon Measures . . . 10
1.12 Hausdorff Measures and Hausdorff Dimensions . . . 13
1.12.1 Hausdorff Measures . . . 13
1.12.2 Hausdorff Dimension . . . 16
1.13 讲义: 几何测度论/ Chapter 2 Orthogonal Projection . . . 17
1.14 Projections of Discrete Sets . . . 17
1.15 Weak Convergence of Measures . . . 20
1.16 Frostman's Lemma . . . 24
1.17 讲义: 几何测度论/ Chapter 3 Differentiation of Measures . . . 26
1.18 Vitali's covering Lemma . . . 26
1.19 Besicovitch's covering lemma . . . 28
1.20 Differentiation of measures . . . 30
1.21 讲义: 几何测度论/ Chapter 4 Fourier Transforms . . . 32
1.22 Schwarz functions and distributions . . . 33
1.23 Fourier transformation of measures . . . 35
1.24 Energy and Fourier transformation . . . 36
1.25 Orthogonal Projections . . . 39
1.26 Falconer's distance set conjecture . . . 40
1.27 Spherical average . . . 43

1.28 讲义: 几何测度论/ Chapter 5 Self-similar Sets 44
1.29 Hausdorff metric 44
1.30 Iterated function system 47
1.31 讲义: 几何测度论/ 习题 52
1.32 Dimensions 52
1.33 Orthogonal Projections 53
1.34 Differentiation of measures 53
1.35 Fourier Transforms 53
1.36 Self-similar sets 54
1.37 试卷: 几何测度论 54

第一章　几何测度论

1.1　习题: 几何测度论

原网页

1.2　Dimensions

The main references of this chapter are [Bishop, Peres; Fractals in Probability and Analysis, Chapter 1], [Evans, Gariepy; Measure Theory and fine properties of functions, Chapter 1 and 2], and [Mattila; Geometry of Sets and Measures in Euclidean Spaces: Fractals and Rectifiability, Chapter 3 and 4].

1.1

1. Let $\mathcal{A} \subset 2^X$ and $\mu : \mathcal{A} \to [0, \infty]$ be such that $\emptyset \in \mathcal{A}$, $X \in \mathcal{A}$, and $\mu(\emptyset) = 0$. For any set $A \subset X$, define
$$\mu^*(A) = \inf\Big\{ \sum_{j=1}^{\infty} \mu(E_j) : E_j \in \mathcal{A} \text{ and } A \subset \cup_{j=1}^{\infty} E_j \Big\}.$$
Show that μ^* is an outer measure.

1.2

1. Let μ be a Borel regular measure on $\mathbb{R}^n$ satisfying $\mu(K) < \infty$ for any compact subset $K \subset \mathbb{R}^n$. Show that Property 1 implies Property 2.
Property 1: If $\mu(B) < \infty$, then, for any $\varepsilon > 0$, there exists a closed set C such that
$$C \subset B, \quad \mu(B \setminus C) < \varepsilon.$$
Property 2: If B is a subset of $\mathbb{R}^n$, then, for any $\varepsilon > 0$, there exists an open set U such that
$$B \subset U, \quad \mu(U \setminus B) < \varepsilon.$$
2. Let $0 < r < \infty$. Show that $\mathcal{H}^n(B(x, r)) \in (0, \infty)$. As a result, there exists $c \in (0, \infty)$ such that $\mathcal{H}^n = c\mathcal{L}^n$.
3. If $f : \mathbb{R}^m \to \mathbb{R}^n$ is Lipschitz, $0 \le s \le m$, and $A \subset \mathbb{R}^m$. Then
$$\mathcal{H}^s(A) \le \mathrm{Lip}(f)^s \mathcal{H}^s(A).$$
In particular, $\dim(fA) \le \dim A$.

1.3 Orthogonal projections

The main references of this chapter are [Folland; Real Analysis, Chapter 4 and Chapter 7], [Evans, Gariepy; Measure theory and fine properties of functions, Chapter 1], [Mattila; Geometry of Sets and Measures in Euclidean Spaces: Fractals and Rectifiability, Chapter 1 and 8], and [Hochman; Lectures on fractal geometry and dynamics, Chapter 7].

2.1

1. Let X be an LCH space, K a compact subset of X, and $\{U_i\}_{i=1}^n$ an open covering of K. There is a partition of unity on K subordinate to $\{U_i\}_{i=1}^n$ consisting of compactly supported functions.

2.2

1. Let μ be an α-Frostman measure on $\mathbb{R}^n$ with $\alpha > n-1$, and $B(0,R) := \{x \in \mathbb{R}^n : |x| \leq R\}$. Show that
$$\mu(\partial B(0,R)) = 0.$$
2. Suppose that μ is an α-Frostman measure. Let $\psi \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ satisfy $\psi \geq 0$ and $\int \psi = 1$. Show that $\mu_\varepsilon := \psi_\varepsilon * \mu$ are α-Frostman measures (uniformly).
3. Suppose that μ is an α-Frostman measure on E. Show that there exist two compact subsets K_1 and K_2 of E with $\operatorname{dist}(K_1, K_2) > 0$ such that $\mu|_{K_1}$ and $\mu|_{K_2}$ are α-Frostman measures.

2.3

1. Let $B \subset \mathbb{R}^n$ be a Borel set of σ-finite $\mathcal{H}^s$-measure. Prove that
$$\mathcal{H}^s(B) = \sup\{\mathcal{H}^s(K) : K \subset B \text{ is a compact set with } \mathcal{H}^s(K) < \infty\}.$$
2. Prove that
$$\int_{\mathbb{R}^n} |f(x)|^p dx = p \int_0^\infty t^{p-1} \left|\{x \in \mathbb{R}^n : |f(x)| > t\}\right| dt.$$
3. Let μ be the measure on $\mathbb{R}^n$ with density $g(x)$. Let $M_\lambda(x) = \lambda x$. Show that $(M_\lambda)_\# \mu$ has density
$$\lambda^{-n} g(\lambda^{-1} x).$$

1.4 Differentiation of measures

The main references of this chapter are [Evans, Gariepy; Measure theory and fine properties of functions, Chapter 1], [Mattila; Geometry of Sets and Measures in Euclidean Spaces: Fractals and Rectifiability, Chapter 2].

3.1

1. Let μ and λ be Radon measures on $\mathbb{R}^n$. Show that $D(\mu, \lambda, x)$ is a Borel function.

2. Let μ and λ be Radon measures on $\mathbb{R}^n$. Suppose that $\mu \ll \lambda$. Show that
$$\int D(\mu,\lambda,x)^2 d\lambda x = \int D(\mu,\lambda,x) d\mu x.$$
3. Let μ and λ be Radon measures on $\mathbb{R}^n$.
   (i) Suppose $\mu \ll \lambda$. Show that
   $$\int_B D(\mu,\lambda,x) d\lambda x = \mu(B).$$
   (ii) Find μ and λ such that
   $$\int_B D(\mu,\lambda,x) d\lambda x < \mu(B)$$
   for some B.

1.5 Fourier Transforms

The main references of this chapter are [Folland; Real Analysis, Chapter 8], [Mattila; Fourier analysis and Hausdorff dimension, Chapter 3, Chapter 4, Chapter 5], and [Mattila 1987; Spherical averages of Fourier transforms of measures with finite energy; dimensions of intersections and distance sets].

4.1

1. A measure is called continuous if each point has measure zero.
   Suppose $\{\mu_N\}, N = 1, 2 \ldots$, and ν are non-negative Radon measures on $\mathbb{R}$, and that ν is continuous. Assume that $\widehat{\mu}_N(\xi) \to \widehat{\nu}(\xi)$ as $N \to \infty$, for each $\xi \in \mathbb{R}$. Then $\mu_N((a,b)) \to \nu((a,b))$ for all $a < b$.
2. [Steinhaus's Theorem] Let E be a measurable subset of $\mathbb{R}$ such that its Lebesgue measure is positive. Then
   $$\Delta'(E) := \{x - y : x, y \in E\}$$
   contains an open interval containing 0.
3. Show that there exist compact sets $A \subset \mathbb{R}^n$ with $\dim A = n$ such that $\Delta(E)$ contains no interval $[0, \varepsilon], \varepsilon > 0$.

4.2

1. Let $\psi \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ satisfy $\int \psi = 1$, $\operatorname{supp}\psi \subset B(0,1)$, and $\psi \geq 0$. Let $\mu \in M(\mathbb{R}^n)$, and $\mu_\varepsilon := \mu * \psi_\varepsilon$. Show that δ_{μ_ε} converges to δ_μ weakly.
2. Let $\mu \in M(\mathbb{R}^n)$. If there exists a constant C such that, for some $s > 0$, we have
   $$\mu(B(x,r)) \geq Cr^S \quad \forall\, x \in \operatorname{supp}\mu, \quad 0 < r < 1.$$
   Show that, for $R > 1$,
   $$\int_{B(0,R)} |\widehat{\mu}(\xi)|^2 d\xi \geq CR^{n-s}.$$

1.6 Self–similar sets

The main references of this chapter are [Hochman; Lectures on fractal geometry and dynamics, Chapter 5], [Bishop, Peres; Fractals in Probability and Analysis, Chapter 2], [Mattila; Fourier analysis and Hausdorff dimension, Chapter 9], and [Shen; Lectures on Bernoulli Convolution].

5.1

1. Let $0 < \varepsilon < \delta$. Show that $A^{(\varepsilon)} \subset A^{(\delta)}$, where $A^{(\varepsilon)} := \{x \in X : d(x, A) < \varepsilon\}$.
2. Let (X, d) be a metric space, and $E \subset X$. Then the following statements are equivalent.
   (i) E is complete and totally bounded.
   (ii) Every sequence in E has a subsequence convergent to a point in E.
   (iii) E is compact.
3. Let $\{\ell_j\}$ be a sequence of lines in $\mathbb{R}^2$ such that
   (i) $\ell_j \cap B(0,1) \neq \emptyset$ for any j,
   and (ii) $\ell'_j := \ell_j \cap B(0,2)$ is convergent under the Hausdorff metric.
   Show that the limit of ℓ'_j is of the form $\ell \cap B(0,2)$ for some line ℓ.

5.2

1. Let Λ be a finite set. For $i, j \in \Lambda^{\mathbb{N}}$, we define $d(i,j) = 2^{-N}$, where N is the largest number such that
   $$i_1 \cdots i_N = j_1 \cdots j_N.$$
   Let τ_d be the topology on $\Lambda^{\mathbb{N}}$ induced by d, and τ the standard product topology on $\Lambda^{\mathbb{N}}$. Show that $\tau = \tau_d$.

5.3

1. Show that the strong separation condition implies the open set condition.
2. Suppose $\Phi = \{\varphi_i\}_{i\in\Lambda}$ is an IFS of similitudes. For $i = (i_1 i_2 \cdots i_n) \in \Lambda^*$, we define $r_i = r_{i_1} r_{i_2} \cdots r_{i_n}$. Fix $r > 0$, and define $S_r = \{i \in \Lambda^* : r_i < r \leq r'_i\}$. Show that S_r is a section.
3. Suppose $(X, \mathcal{A}, \mu)$ is a measurable space, and we call $A \in \mathcal{A}$ a μ-atom if $\mu(A) > 0$, and for any $B \in \mathcal{A}$ with $B \subset A$ we have either $\mu(B) = 0$ or $\mu(A \setminus B) = 0$. Show that
   (i) Let A be a μ-atom. Then for any $B \in \mathcal{A}$ with $B \setminus A$ we have either $\mu(B) = \mu(A)$ or $\mu(B) = 0$.
   (ii) When $X = \mathbb{R}^n$, and μ is a Borel measure, show that a μ-atom is a singleton $\{x\}$ with $\mu(\{x\}) > 0$.

1.7 讲义: 几何测度论

原网页

讲义目前大修中。若在此期间发现讲义有明显错误，请耐心等待。

本讲义是根据贺丹青老师 2025 年秋季的课程整理而成。若有错漏，多为整理者笔误或 AI 在调整 markdown 格式笔记为 TeX 格式时出错，敬请谅解。也可联系整理者 (周铭杨, 邮箱：myzhou25@m.fudan.edu.cn) 获取.md 格式的笔记。

- Chapter 1 Measure Theory
- Chapter 2 Orthogonal Projection
- Chapter 3 Differentiation of Measures
- Chapter 4 Fourier Transforms
- Chapter 5 Self-similar Sets
- 习题

1.8　讲义: 几何测度论/ Chapter 1 Measure Theory

原网页

The main references of this chapter are [Bishop, Peres; Fractals in Probability and Analysis, Chapter 1], [Evans, Gariepy; Measure theory and fine properties of functions, Chapter 1 and 2], and [Mattila; Geometry of Sets and Measures in Euclidean Spaces-Fractals and Rectifiability, Chapter 3 and 4].

1.9　Middle-α Cantor Sets

Let $\alpha \in (0,1)$, and let C_α be the middle-α Cantor set.

Goal. *Construct $C_{\alpha,n}$, which is a union of 2^n disjoint closed intervals, indexed by $i = i_1, i_2, \ldots, i_n \in \{0,1\}^n$, each of length $\left(\frac{1-\alpha}{2}\right)^n$, such that $C_\alpha = \bigcap_{n=0}^{\infty} C_{\alpha,n}$.*

例 1.9.1. *Classical Cantor set $\alpha = \frac{1}{3}$, $I = [0,1]$.*

$$
\begin{aligned}
&C_{\frac{1}{3},0} := I \\
&C_{\frac{1}{3},1} := \left[0, \frac{1}{3}\right] \cup \left[\frac{2}{3}, 1\right] \\
&C_{\frac{1}{3},2} := \left[0, \frac{1}{9}\right] \cup \left[\frac{2}{9}, \frac{1}{3}\right] \cup \left[\frac{2}{3}, \frac{7}{9}\right] \cup \left[\frac{8}{9}, 1\right] \\
&\left(\textit{Index}: \frac{2}{3}\cdot 0 + \frac{2}{9}\cdot 0, \frac{2}{3}\cdot 0 + \frac{2}{9}\cdot 1, \frac{2}{3}\cdot 1 + \frac{2}{9}\cdot 0, \frac{2}{3}\cdot 1 + \frac{2}{9}\cdot 1 \textit{ indicate where these intervals begin}\right) \\
&\quad \cdots \\
&C_{\frac{1}{3},n} := \bigcup_{i_1,i_2,\ldots,i_n \in \{0,1\}} \left[\sum_{k=1}^{n} \frac{2}{3^k} i_k, \sum_{k=1}^{n} \frac{2}{3^k} i_k + \left(\frac{1}{3}\right)^n\right] \\
&C_{\alpha,n} := \bigcup_{i \in \{0,1\}^n} I_i.
\end{aligned}
$$

By removing from I_i the open interval with the same center as I_i and length $\alpha\,|I_i|$, we obtain two

subintervals $I_{i_1 i_2 \cdots i_n 0}$ and $I_{i_1 i_2 \cdots i_n 1}$.

$$\begin{aligned} C_{\alpha,n+1} &:= \bigcup_{i\in\{0,1\}^n} (I_{i0} \cup I_{i1}) \\ &= \bigcup_{i\in\{0,1\}^{n+1}} I_i. \end{aligned}$$

Clearly, $C_{\alpha,n+1} \subset C_{\alpha,n}$, and $\forall k, C_{\alpha,k}$ is compact. Thus $C_\alpha = \bigcap_{n=0}^{\infty} C_{\alpha,n}$ is compact and non-empty.

注 1.9.1. *(i)* $\mathcal{L}^1(C_\alpha) = 0, \forall \alpha \in (0,1)$*;*

(ii) C_α *are mutually homeomorphic.*

From the remark above we know that the Lebesgue measure is not a good tool to measure the size of Cantor sets. We will introduce Minkovski dimension to study the size of C_α.

1.10 Minkovski Dimension

By talking about dimension, we want to know how a set scales. For example, a line segment of length L can be covered by $\frac{L}{\epsilon}$ intervals of length ϵ. A square of side length L can be covered by $\left(\frac{L}{\epsilon}\right)^2$ squares of side length ϵ. A cube of side length L can be covered by $\left(\frac{L}{\epsilon}\right)^3$ cubes of side length ϵ. Thus we have the following definition.

定义 1.10.1. *(i)* $\forall E \subset \mathbb{R}^n$*, the diameter of* E *is defined as* $\operatorname{diam} E := \sup_{x,y\in E} |x-y|$.

(ii) If $E \subset \bigcup_{\alpha\in A} E_\alpha$ *then* $\{E_\alpha\}_{\alpha\in A}$ *is a covering of* E*. If* $\{E_\alpha\}_{\alpha\in A}$ *also satisfies* $\operatorname{diam} E_\alpha \le \delta, \forall \alpha \in A$*, then it is called a* δ*-covering of* E*.*

定义 1.10.2. *(i) Given a bounded set* $K \subset \mathbb{R}^n$*, let* $N(K,\delta)$ *be the minimal cardinality of a* δ*-covering of* K*, i.e.,* $N(K,\delta) = \min\left\{J : K \subset \bigcup_{j=1}^{J} E_j, \operatorname{dim}(E_j) \le \delta\right\}$.

(ii) The upper Minkovski dimension of a bounded set K *is*

$$\overline{\dim}_M(K) := \limsup_{\delta\to 0} \frac{\log N(K,\delta)}{-\log \delta}.$$

The lower Minkovski dimension of a bounded set K *is*

$$\underline{\dim}_M(K) := \liminf_{\delta\to 0} \frac{\log N(K,\delta)}{-\log \delta}.$$

(iii) If $\overline{\dim}_M(K) = \underline{\dim}_M(K)$*, then we call it the Minkovski dimension of* K*, denoted by* $\dim_M(K)$.

注 1.10.1. *(i)* $\dim_M(K) = \alpha$ *iff* $\forall \epsilon > 0, \exists \delta_0 > 0$*, s.t.*

$$\delta^{-\alpha-\epsilon} \le N(K,\delta) \le \delta^{-\alpha+\epsilon}, \forall \delta \in (0,\delta_0).$$

(ii) $\underline{\dim}_M(K) = \alpha$ *iff* $\forall \epsilon > 0, \exists \delta_0 > 0$*, s.t.*

$$-\alpha - \epsilon \le \inf_{\eta\in(0,\delta)} \frac{\log N(K,\eta)}{\log \eta^{-1}} \le -\alpha + \epsilon, \forall \delta \in (0,\delta_0).$$

(iii) $E \subset F \Leftrightarrow \dim_M E \le \dim_M F$.

引理 1.10.1. *(i) Let $N_B(K,\delta)$ be the minimal cardinality of a covering of E consisting of balls of radius δ. Then $\dim_M(K) = \lim_{\delta\to 0} \frac{\log N_B(K,\delta)}{-\log\delta}$ whenever $\dim_M(K)$ exists.*

(ii) In (i), it is equivalent to use axis-aligned cubes of side length δ to cover K.

(iii) If $A \subset \mathbb{R}^n$ is bounded, then $\overline{\dim}_M(A) = \overline{\dim}_M(\bar{A})$, $\underline{\dim}_M(A) = \underline{\dim}_M(\bar{A})$.

证明. (i) Let $N_B(K,\delta)$ be the minimal cardinality of a covering of E consisting of balls of radius δ. Then any such covering is also a δ-covering, so $N(K,\delta) \le N_B(K,\delta)$. On the other hand, if $\{E_j\}_{j=1}^{N(K,\delta)}$ is a δ-covering of K, then $\{B(x_j,\delta)\}_{j=1}^{N(K,\delta)}$ is a covering of K, where $x_j \in E_j$. Thus $N_B(K,\delta) \le N(K,\delta)$. Hence $N(K,\delta) = N_B(K,\delta)$, and the result follows.

(ii) Similar to (i).

(iii) We fix a covering $\{E_j\}_{j=1}^{J}$ of A with $B_j = B(a_j,\delta)$ satisfying $J = N_B(A,\delta)$. As J is finite, we have

$$\bar{A}_j \subseteq \overline{\bigcup_{j=1}^{J} B_j} = \bigcup_{j=1}^{J} \overline{B_j} = \bigcup_{j=1}^{J} B_j. \qquad \square$$

例 1.10.1. *(i) If E is finite, then $\dim_M(E) = 0$.*

(ii) If E is a line segment in $\mathbb{R}^n$, then $\dim_M(E) = 1$. More generally, any compact k-dimensional C^1 submanifold of $\mathbb{R}^n$ has Minkovski dimension k.

(iii) Let $0 < \alpha < 1$, then $\dim_M(C_\alpha) = \frac{\log 2}{\log 2 - \log(1-\alpha)}$.

(iv) If $E \subseteq \mathbb{R}^n$ and $\dim_M E < n$, then $\mathcal{L}^n(E) = 0$.

(v) Let $E = \{0\} \cup \{1, \frac{1}{2}, \frac{1}{3}, \ldots\}$, then $\dim_M E = \frac{1}{2}$.

(Idea: cover the first n points from 1, then cover the rest points from $\frac{1}{n}$ by n intervals of length $\frac{1}{n^2}$.)

注 1.10.2. *From (v) we know that Minkovski dimension is not countably stable, i.e., $\dim_M(\bigcup_{i=1}^{\infty} E_i) \neq \sup_i \dim_M(E_i)$ in general.*

For a given set $E \subset \mathbb{R}^n$, we define the δ-neighborhood of E as

$$E_\delta := \{y \in \mathbb{R}^n : \operatorname{dist}(y,E) < \delta\}.$$

引理 1.10.2. *Let $\alpha > 0$, suppose that $\exists A_\alpha > 0$ s.t. $\mathcal{L}(E_\delta) \ge A_\alpha \delta^\alpha$, $\forall \delta > 0$. Then $\underline{\dim}_M(E) \ge n - \alpha$.*

证明. Suppose $E \subseteq \bigcup_{j=1}^{J} B(x_j,\delta)$, then $E_\delta \subseteq \bigcup_{j=1}^{J} B(x_j,2\delta)$, which implies that $A_\alpha\delta^\alpha \le \mathcal{L}(E_\delta) \le C_n J \delta^n$. As a result, $N_B(E,\delta) \ge A_{\alpha,n}\delta^{-(n-\alpha)}$. Therefore $\underline{\dim}_M(E) \ge n - \alpha$. $\square$

1.11 Measures

Let X be a non-empty set. Denote by 2^X be the power set of X, i.e., the set of all subsets of X.

定义 1.11.1. *A map $\mu : 2^X \to [0,+\infty]$ is called a measure on X if*

(i) $\mu(\emptyset) = 0$;

(ii) (Countable subadditivity) If $A \subseteq \bigcup_{k=1}^{\infty} A_k$, then $\mu(A) \le \sum_{k=1}^{\infty} \mu(A_k)$.

注 1.11.1. *In most references, the measure as defined above is called an outer measure.*

定义 1.11.2. *Let μ be a measure on X. C is a subset of X. Define the restriction of μ to C as*

$$\mu|_C(A) := \mu(A \cap C), \forall A \subseteq X.$$

定义 1.11.3. *A set $A \subseteq X$ is called μ-measurable if $\forall B \subseteq X$,*

$$\mu(B) = \mu(B \cap A) + \mu(B \cap A^c).$$

定理 1.11.1. *Let μ be a measure on X.*

(i) If $A \subseteq B \subseteq X$, then $\mu(A) \leq \mu(B) \leq \mu(X)$.

(ii) If $A \subseteq X$ is μ-measurable, then so is A^c.

(iii) $\varnothing$ and X are μ-measurable. Moreover, if $\mu(A) = 0$, then A is μ-measurable.

(iv) Let $C \subseteq X$. Then any μ-measurable set is also $\mu|_C$-measurable.

证明.　(ii) $\mu(B) = \mu(B \cap A) + \mu(B \setminus A) = \mu(B \setminus A^C) + \mu(B \cap A^C)$.

(iv) $\mu|_C(B) = \mu(B \cap C) = \mu((B \cap C) \cap A) + \mu((B \cap C) \cap A^c) = \mu|_C(B \cap A) + \mu|_C(B \cap A^c)$. □

定理 1.11.2. *Let $\{A_k\}_{k=1}^{\infty}$ be a sequence of μ-measurable sets.*

(i) If A_k are pairwise disjoint, then $\mu(\bigcup_{k=1}^{\infty} A_k) = \sum_{k=1}^{\infty} \mu(A_k)$.

(ii) If $A_1 \subseteq \cdots \subseteq A_k \subseteq A_{k+1} \subseteq \cdots$, then

$$\lim_{k\to\infty} \mu(A_k) = \mu\left(\bigcup_{k=1}^{\infty} A_k\right).$$

(iii) If $A_1 \supseteq \cdots \supseteq A_k \supseteq A_{k+1} \supseteq \cdots$ and $\mu(A_1) < +\infty$, then

$$\lim_{k\to\infty} \mu(A_k) = \mu\left(\bigcap_{k=1}^{\infty} A_k\right).$$

(iv) $\bigcup_{k=1}^{\infty} A_k$ and $\bigcap_{k=1}^{\infty} A_k$ are μ-measurable.

证明. First we prove that for finie n, $\bigcup_{k=1}^{n} A_k$ is measurable. By induction, we only need to prove the case $n = 2$. For any $B \subseteq X$,

$$\begin{aligned}\mu(B) &= \mu(B \cap A_1) + \mu(B \cap A_1^C) \\ &= \mu(B \cap A_1) + \mu(B \cap A_1^C \cap A_2) + \mu(B \cap A_1^C \cap A_2^C) \\ &\geq \mu(B \cap (A_1 \cup A_2)) + \mu(B \cap (A_1 \cup A_2)^C).\end{aligned}$$

(Note that $A_1 \cup A_2 = A_1 \cup (A_1^C \cap A_2)$.) Thus $A_1 \cup A_2$ is measurable. Now $(A_1 \cap A_2)^C = A_1^C \cup A_2^C$, so $A_1 \cap A_2$ is also measurable.

(i) Let $B_j = \bigcup_{k=1}^{j} A_k$. Then

$$\begin{aligned}\mu(B_{j+1}) &= \mu(B_{j+1} \cap A_{j+1}) + \mu(B_{j+1} \cap A_{j+1}^C) \\ &= \mu(A_{j+1}) + \mu(B_j),\end{aligned}$$

therefore $\mu(\bigcup_{k=1}^{\infty} A_k) \geq \mu(\bigcup_{k=1}^{j} A_k) = \sum_{k=1}^{j} \mu(A_k)$. By letting $j \to \infty$, we obtain the result from subadditivity.

(ii) From (i) we know

$$\begin{aligned}\lim_{k\to\infty}\mu(A_k) &= \mu(A_1)+\sum_{k=1}^{\infty}(\mu(A_{k+1})-\mu(A_k))\\ &= \mu(A_1)+\sum_{k=1}^{\infty}\mu(A_{k+1}\setminus A_k)\\ &= \mu\left(\bigcup_{k=1}^{\infty}A_k\right).\end{aligned}$$

(iii) Let $B_k = A_1 \setminus A_k$, then by (ii) we have

$$\begin{aligned}\lim_{k\to\infty}\mu(B_k) = \mu(\bigcup_{k=1}^{\infty}B_k) &= \mu(A_1\setminus\bigcap_{k=1}^{\infty}A_k)\\ \overset{\mu(A_1)<+\infty}{\Longleftrightarrow}\ \lim_{k\to\infty}\mu(A_k) &= \mu\left(\bigcap_{k=1}^{\infty}A_k\right).\end{aligned}$$

(iv) Since $B_j = \bigcup_{k=1}^{j}A_k$ is measurable, for any $B\subseteq X$ with $\mu(B)<\infty$, we have

$$\begin{aligned}\mu(B\cap\bigcup_{k=1}^{\infty}A_k)+\mu(B\setminus\bigcup_{k=1}^{\infty}A_k) &= \mu|_B(\bigcup_{k=1}^{\infty}A_k)+\mu|_B(\bigcap_{k=1}^{\infty}A_k)\\ &= \lim_{k\to\infty}\left[\mu|_B(B_k)+\mu|_B(B_k^C)\right]=\mu(B).\end{aligned}$$

That is, $\bigcup_{k=1}^{\infty}A_k$ is μ-measurable. $\left(\bigcap_{k=1}^{\infty}A_k\right)^C = \bigcup_{k=1}^{\infty}A_k^C \Rightarrow \bigcap_{k=1}^{\infty}A_k$ is also μ-measurable. □

定义 1.11.4. *A collection of subsets $\mathscr{A}\subset 2^X$ is called a σ-algebra if*

(i) $\varnothing, X\in\mathscr{A}$;

(ii) $A\in\mathscr{A}\Rightarrow A^c := X\setminus A\in\mathscr{A}$;

(iii) $A_k\in\mathscr{A}, k\in\mathbb{Z}_+\Rightarrow\bigcup_{k=1}^{\infty}A_k\in\mathscr{A}$.

注 1.11.2. $\bigcap_{k=1}^{\infty}A_k = (\bigcup_{k=1}^{\infty}A_k^c)^c\in\mathscr{A}$.

定理 1.11.3. *Let μ be a measure on a nonempty set X. Then the collection M of all μ-measurable subsets of X is a σ-algebra.*

注 1.11.3. *By Theorem 1.11.2(i), μ is countably additive on M.*

定义 1.11.5. *If $\mathcal{C}\subset 2^X$ is a nonempty collection of subsets of X, the σ-algebra generated by $\mathcal{C}$, denoted by $\sigma(\mathcal{C})$, is the smallest σ-algebra containing $\mathcal{C}$.*

注 1.11.4. *$\sigma(\mathcal{C})$ is the intersection of all σ-algebra containing $\mathcal{C}$.*

定义 1.11.6. *(i) The Borel σ-algebra of $\mathbb{R}^n$ is the smallest σ-algebra $\mathscr{B}$ of $\mathbb{R}^n$ containing all open subsets of $\mathbb{R}^n$. An element in $\mathscr{B}$ is called a Borel set.*

(ii) A measure μ is called Borel if every Borel set is μ-measurable.

定理 1.11.4 (Caratheodory's criterion)**.** *Let μ be a measure on $\mathbb{R}^n$. If for all sets $A, B\subset\mathbb{R}^n$, we have*

$$\mu(A\cup B)=\mu(A)+\mu(B)\quad \textit{whenever } \operatorname{dist}(A,B)>0 \tag{1.1}$$

then μ is a Borel measure.

证明. It suffices to show that a closed set C is μ-measurable, which follows from the inequality

$$\mu(A) \geq \mu(A \cap C) + \mu(A \setminus C), \forall A \subset \mathbb{R}^n. \tag{1.2}$$

We may assume $\mu(A) < \infty$, or otherwise (1.2) is trivial.

Define for all $n \in \mathbb{Z}_+$

$$C_n := \left\{ x \in \mathbb{R}^n : \text{dist}(x, C) \leq \frac{1}{n} \right\},$$

then $\text{dist}(A \cap C, A \cap C_n^c) \geq \frac{1}{n} > 0$.

By (1.1),

$$\mu(A \cap C) + \mu(A \cap C_n^c) = \mu((A \cap C) \cup (A \cap C_n^c)),$$

we complete the proof if we have

$$\lim_{n \to \infty} \mu(A \cap C_n^c) = \mu(A \cap C^c). \tag{1.3}$$

For $k \in \mathbb{Z}_+$, define

$$R_k := \{x \in A : (k+1)^{-1} < \text{dist}(x, C) \leq k^{-1}\}.$$

Observe that $\text{dist}(R_i, R_j) > 0$ if $j \geq i + 2$, therefore, by (1.1),

$$\sum_{k=1}^{j} \mu(R_{2k}) = \mu(\bigcup_{k=1}^{j} R_{2k}) \leq \mu(A).$$

Taking $j \to \infty$, and considering the odd terms, we have

$$\sum_{k=1}^{\infty} \mu(R_k) \leq 2\mu(A) < \infty.$$

In particular,

$$\lim_{n \to \infty} \sum_{k=n}^{\infty} \mu(R_k) = 0.$$

Recalling that C is closed, $\text{dist}(x, C) > 0, \forall x \in C^c$, which implies that

$$A \cap C^c = (A \cap C_n^c) \cup \left(\bigcup_{k=n}^{\infty} R_k \right).$$

As a result,

$$\mu(A \cap C_n^c) \leq \mu(A \cap C) \leq \mu(A \cap C_n^c) + \sum_{k=n}^{\infty} \mu(R_k).$$

Taking $n \to \infty$, we obtain (1.3). □

1.11.1 Radon Measures

定义 1.11.7. *Let μ be a measure on X.*

(i) μ is locally finite if $\forall x \in X, \exists r > 0$ s.t. $\mu(B(x, r)) < \infty$.

(ii) μ is Borel regular if it is a Borel measure and $\forall A \subset X, \exists$ a Borel set $B \subset X$ s.t. $A \subset B$, and $\mu(A) = \mu(B)$.

(iii) μ *is a Radon measure if it is Borel regular and locally finite.*

注 1.11.5. *(i) On* $\mathbb{R}^n$*,* μ *is locally finite iff* $\mu(K) < \infty$ *for all compact* K*.*

(ii) The initial definition of measure does not relate to topology, but Radon measures do.

定理 1.11.5. *Let* μ *be a Borel regular measure on* $\mathbb{R}^n$*. Suppose* $A \subset \mathbb{R}^n$ *is* μ*-measurable and* $\mu(A) < \infty$*. Then* $\mu|_A$ *is a Radon measure.*

注 1.11.6. *If* A *is a Borel set, then* $\mu|_A$ *is Borel regular even if* $\mu(A) = \infty$*.*

证明. Denote $\nu := \mu|_A$. Obviously, $\nu(K) < \infty$ for all compact K. As every μ-measurable set is ν-measurable, and μ is Borel, we have that ν is Borel.

It remains to show that $\forall C \subset \mathbb{R}^n, \exists$ a Borel set $B \supset C$ s.t. $\nu(B) = \nu(C)$.

We may assume that A is a Borel set. In fact, $\exists$ a Borel set B s.t. $A \subset B$ and $\mu(A) = \mu(B)$ as μ is regular. Which implies that

$$\begin{aligned}\mu|_B(C) &= \mu(B \cap C \cap A) + \mu(B \cap C \cap A^c) \\ &\leq \mu(A \cap C) + \mu(B \cap A^c) \\ &= \mu(A \cap C) + \mu(B) - \mu(A) \\ &= \mu|_A(C)\end{aligned}$$

as A is measurable, and $\mu(A) < \infty$.

Fix $C \subset \mathbb{R}^n$, then $\exists$ a Borel set E s.t. $A \cap C \subset E$, and $\mu(A \cap C) = \mu(E)$. Let $D := E \cup A^c$, which is Borel. Then $C \subset (A \cap C) \cup A^c \subset D$, and $\nu(D) = \mu(D \cap A) \leq \mu(E \cap A) \leq \mu(E) = \nu(C)$. Hence $\nu(C) = \nu(D)$. □

注 1.11.7. *Hausdorff measure is infinite on lower dimensional subspaces. But Radon measure requires a finite value on compact sets. Therefore, most Hausdorff measures are not Radon measures.*

引理 1.11.1. *Let* μ *be a Borel measure on* $\mathbb{R}^n$*, and let* B *be a Borel set.*

(i) If $\mu(B) < \infty$*, then* $\forall \epsilon > 0, \exists$ *a closed set* C *s.t.* $C \subset B$ *and* $\mu(B \setminus C) < \epsilon$*.*

(ii) If μ *is a Radon measure, then* $\forall \epsilon > 0$*,* $\exists$ *an open set* U *s.t.* $B \subset U$*, and* $\mu(U \setminus B) < \epsilon$*.*

证明. Let $\nu := \mu|_B$, which is a finite Borel measure. Let

$$\mathscr{F} := \{A \subset \mathbb{R}^n : A \text{ is } \mu\text{-measurable, and } \forall \epsilon > 0, \exists \text{ closed } C \text{ s.t. } \nu(A \setminus C) < \epsilon\}.$$

Obviously, all closed sets are in $\mathscr{F}$. We will show that

$$\mathscr{G} := \{A \in \mathscr{F} : \mathbb{R}^n \setminus A \in \mathscr{F}\}$$

is a σ-algebra containing all Borel sets. In particular, $B \in \mathscr{G}$, i.e., (i) holds.

Claim 1 If $\{A_i\}_{i=1}^{\infty} \subset \mathcal{F}$, then $A = \bigcap_{i=1}^{\infty} A_i \in \mathscr{F}$

Proof of Claim 1: Fix $\epsilon > 0$. Since $A_i \in \mathscr{F}$, $\exists$ a closed set $C_i \subset A_i$ s.t. $\nu(A_i \setminus C_i) < 2^{-i-1}\epsilon$. Let $C = \bigcap_{i=1}^{\infty} C_i$, then C is closed, and

$$\nu(A \setminus C) \leq \nu\left(\bigcup_{i=1}^{\infty}(C_i^c \cap A_i)\right) < \epsilon.$$

Claim 2 If $\{A_i\}_{i=1}^{\infty} \subset \mathscr{F}$, then $A := \bigcup_{i=1}^{\infty} A_i \subset \mathscr{F}$.

Proof of Claim 2: As $\nu(A) < \infty$, we have

$$\lim_{m\to\infty} \nu\left(A \setminus \bigcup_{i=1}^{m} C_i\right) \leq \nu\left(\bigcup_{i=1}^{\infty}(A_i \cap C_i^c)\right) < \frac{\epsilon}{2}.$$

Consequently, $\exists m$ s.t. $\mu\left(A \setminus \bigcup_{i=1}^{m} C_i\right) < \epsilon$, as $\bigcup_{i=1}^{m} C_i$ is closed, $A \in \mathscr{F}$.

As every open set in $\mathbb{R}^n$ can be written as a countable union of closed sets, we obtain from Claim 2 that $\mathscr{F}$ contains all open sets. By definition, $\mathscr{G}$ contains all open sets. Suppose $\{A_i\}_{i=1}^{\infty} \subset \mathscr{G}$, then $A_i, A_i^c \in \mathscr{F}$. By Claim 1 and 2, $\bigcup_{i=1}^{\infty} A_i, \bigcap_{i=1}^{\infty} A_i^c \in \mathscr{F}$, hence $\bigcup_{i=1}^{\infty} A_i \in \mathscr{G}$. As a result, $\mathscr{G}$ is a σ-algebra containing all Borel sets.

For (ii): Let $U_m := B(0,m) = \{x \in \mathbb{R}^n : |x| \leq m\}$. As μ is Radon, we have $\mu(U_m \setminus B) < \infty$. Applying (i), we find a closed set $C_m \subset U_m \cap B^c$ s.t. $\mu(U_m \cap B^c \cap C_m^c) < 2^{-m}\epsilon$. Let $U := \bigcup_{m=1}^{\infty}(U_m \cup C_m^c)$, which is open. As $C_m \subset B^c$, we have $B \subset C_m^c$, and therefore $U_m \cap B \subset U_m \cap C_m^c$. Consequently, $B \subset U$. Furthermore, $\mu(U \setminus B) \leq \sum_m \mu(U_m \cap C_m^c \cap B^c) < \epsilon$. □

定理 1.11.6. *Let μ be a Radon measure on $\mathbb{R}^n$. Then*

(i) $\forall A \subset \mathbb{R}^n$,

$$\mu(A) = \inf\{\mu(U) : A \subset U, U \text{ is open}\} \tag{1.4}$$

(inner regularity)

(ii) $\forall \mu$*-measurable set* $A \subset \mathbb{R}^n$,

$$\mu(A) = \sup\{\mu(K) : K \subset A, K \text{ is compact}\}$$

(outer regularity)

注 1.11.8. *In many references this is taken as the definition of Radon measure, but geometric measure theory often uses the definition above.*

证明. The conclusion is obvious if $\mu(A) = \infty$, so we assume $\mu(A) < \infty$.

Suppose A is Borel. Fixing $\epsilon > 0$, by Lemma above, $\exists$ an open set $U \supset A$ s.t. $\mu(U \setminus A) < \epsilon$. As A is μ-measurable, we have $\mu(U) = \mu(A) + \mu(U \cap A^c)$, which implies (1.4).

For an arbitrary set A, $\exists$ a Borel set B s.t. $A \subset B$ and $\mu(A) = \mu(B)$. Then

$$\mu(A) = \mu(B) = \inf\{\mu(U) : B \subset U, U \text{ is open}\} \geq \inf\{\mu(U) : A \subset U, U \text{ is open}\}.$$

The opposite inequality is trivial, so (i) holds.

For (ii): Consider first the case $\mu(A) < \infty$. Let $\nu := \mu|_A$, which is Radon by previous theorem. Fixing $\epsilon > 0$, and applying (i) to ν and A^c, $\exists$ an open set $U \supset A^c$ s.t. $\nu(U) \leq \epsilon$. Denote $C = U^c$, then C is closed, and $C \subset A$. Moreover, $\mu(A \setminus C) = \nu(C^c) \leq \epsilon$. Thus, $0 \leq \mu(A) - \mu(C) \leq \epsilon$, and therefore $\mu(A) = \sup\{\mu(C) : C \subset A, C \text{ is closed}\}$.

Let $C_m := C \cap \overline{B(0,m)}$, then C_m is compact, and $\lim_{m\to\infty} \mu(C_m) = \mu(C)$. Then (ii) holds when $\mu(A) < \infty$.

Now suppose $\mu(A) = \infty$. Define $D_k := \{x \in \mathbb{R}^n : k-1 \leq |x| < k\}$. Then $A = \bigcup_{k=1}^{\infty}(D_k \cap A)$, which implies that $\infty = \mu(A) = \sum_{k=1}^{\infty} \mu(D_k \cap A)$. Since μ is Radon, $\mu(D_k \cap A) < \infty$. Then $\exists$ a compact

set $C_k \subset D_k \cap A$ s.t. $\mu(C_k) \geq \mu(D_k \cap A) - 2^{-k-1}$. As a result, $\bigcup_{k=1}^{\infty} C_k \subset A$, $\lim_{m\to\infty} (\bigcup_{k=1}^{\infty} C_k) = \mu(\bigcup_{k=1}^{\infty} C_k) = \sum \mu(C_k) = \infty$. This yields (ii) as $\bigcup_{k=1}^{m} C_k$ is compact. □

定义 1.11.8. *(i) A topological space X is called locally compact if every point has a compact neighborhood.*

(ii) X is called σ-compact if it is a countable union of compact sets.

定理 1.11.7. *Let X be a locally compact Hausdorff space in which every open set is σ-compact. Then every Borel measure on X that is finite on compact sets is Radon. (Proof omitted.)*

1.12 Hausdorff Measures and Hausdorff Dimensions

1.12.1 Hausdorff Measures

定义 1.12.1. *(i) Let $A \subset \mathbb{R}^n$, $0 \leq s < \infty$, $0 < \delta \leq \infty$. Define*

$$H^s_\delta(A) := \inf\left\{\sum_{j=1}^{\infty} \alpha(s)\left(\frac{\operatorname{diam} C_j}{2}\right)^s : A \subset \bigcup_{j=1}^{\infty} C_j, \operatorname{diam} C_j \leq \delta\right\},$$

where

$$\alpha(s) := \frac{\pi^{s/2}}{\Gamma\left(\frac{s}{2}+1\right)}.$$

(ii) 定义

$$H^s(A) := \lim_{\delta\to 0} H^s_\delta(A) = \sup_{\delta>0} H^s_\delta(A).$$

注 1.12.1. $\alpha(s)$ 是一个与球体积有关的常数，是为了使 s 取整时 *Hausdorff* 测度与经典 *Lebesgue* 测度相等。其并不会对 *Hausdorff* 维数产生本性影响。

注 1.12.2. *(i)* $\Gamma(s) := \int_0^\infty e^{-x}x^{s-1}\mathrm{d}x$.

(ii) $\alpha(n) = \mathcal{L}^n(B(0,1))$.

(iii) $H^n = \mathcal{L}^n$.

(iv) The definition remains equivalent if we assume that C_j are closed balls.

定义 1.12.2. *(i) Let $A \subset \mathbb{R}^n$, $0 \leq s < \infty$, $0 < \delta \leq \infty$. Define*

$$H^s_\delta(A) := \inf\left\{\sum_{i=1}^{\infty} \alpha(s)\left(\frac{\operatorname{diam} C_j}{2}\right)^s : A \subset \bigcup_{j=1}^{\infty} C_j, \operatorname{diam} C_j \leq \delta\right\},$$

where $\alpha(s)$ is as above.

(ii) Define

$$H^s(A) := \lim_{\delta\to 0} H^s_\delta(A) = \sup_{\delta>0} H^s_\delta(A).$$

定理 1.12.1. *For all $0 \leq s < n$, H^s is a Borel regular measure in $\mathbb{R}^n$.*

证明. **Step 1** H^s_δ is a measure. Clearly $H^s_\delta \neq 0$. Suppose $A \subset \bigcup_{j=1}^\infty A_j$. Take a δ-covering $\{C_j^k\}_{k=1}^\infty$ of A_j for each j, then $\{C_j^k\}_{j,k=1,2,\dots}$ is a δ-covering of A. This implies

$$H^s_\delta(A) \leq \sum_{j=1}^\infty \sum_{k=1}^\infty (\operatorname{diam} C_j^k)^s$$

Taking the infimum, we get

$$H^s_\delta(A) \leq \sum_{j=1}^\infty H^s_\delta(A_j). \tag{1.5}$$

Step 2 H^s is a measure. By (1.5), $H^s_\delta(A) \leq \sum_{j=1}^\infty H^s(A_j)$. Letting $\delta \to 0$, we obtain

$$H^s(A) \leq \sum_j H^s(A_j).$$

Step 3 H^s is a Borel measure. By Caratheodory's criterion (see Theorem 1.11.4), it suffices to show that if $\operatorname{dist}(A, B) > 0$, then

$$H^s(A \cup B) = H^s(A) + H^s(B).$$

Let $\delta < \frac{1}{4} \operatorname{dist}(A, B)$. Take a δ-covering $\{C_j\}$ of $A \cup B$. Then

$$\sum_{j=1}^\infty (\operatorname{diam} C_j)^s \geq \sum_{C_j \cap A \neq \varnothing} (\operatorname{diam}(C_j)^s) + \sum_{C_j \cap B \neq \varnothing} (\operatorname{diam}(C_j)^s) \geq H^s_\delta(A) + H^s_\delta(B).$$

Taking infimum and letting $\delta \to 0$, we have $H^s(A \cup B) \geq H^s(A) + H^s(B)$. Furthermore, $H^s(A \cup B) = H^s(A) + H^s(B)$ since H^s is a measure.

Step 4 H^s is Borel regular. Fix $A \subset \mathbb{R}^n$. It suffices to show that there exists a Borel set $B \supset A$ such that $H^s(A) = H^s(B)$. WLOG $H^s(A) < \infty$. By definition, there exists a k^{-1}-covering $\{C_j^k\}$ of A such that

$$\sum_{j=1}^\infty (\operatorname{diam} C_j^k)^s \leq H^s(A) + k^{-1}.$$

As $\operatorname{diam} C = \operatorname{diam} \bar{C}$, we may assume that C_j^k are closed. Let $B := \bigcap_{k=1}^\infty \bigcup_{j=1}^\infty C_j^k$, then B is a Borel set containing A, and

$$H^s_{k^{-1}}(B) \leq H^s(A) + k^{-1}.$$

Letting $k \to \infty$, we have $H^s(B) = H^s(A)$. □

定理 1.12.2. *(i) For $0 \leq s < n$, H^s is not locally finite. In particular, H^s is not a Radon measure.*

(ii) Let $A \subset \mathbb{R}^n$ be H^s-measurable, and $H^s(A) < \infty$. Then $H^s|_A$ is a Radon measure.

(iii) H^n on $\mathbb{R}^n$ is a Radon measure.

(iv) For all $s > n$, $H^s \equiv 0$ on $\mathbb{R}^n$.

证明. (i) Let $B := \{x \in \mathbb{R}^n : |x| \leq 1\}$. It suffices to show that $H^s(B) = \infty$ for $0 \leq s < n$. To cover B using balls of radius δ, we need at least $C_n \delta^{-n}$ such balls. This yields

$$H^s_\delta(B) \geq C_{s,n} \delta^{-n} \delta^s$$

Letting $\delta \to 0$, we obtain $H^s(B) = \infty$ since $s < n$.

(ii) Apply Theorem 1.12.1 and Theorem [原站引用：thm:radon-restriction].

(iii) It suffices to show that $H^n(B) < \infty$. For any $\delta > 0$, we can find $C_n\delta^{-n}$ balls of radius δ to cover B, then

$$H^n_\delta(B) \leq C'_n$$

This implies $H^n(B) \leq C'_n < \infty$.

(iv) Obvious. □

定理 1.12.3. *(i) H^0 is the counting measure.*

(ii) $H^s(\lambda A) = \lambda^s H^s(A)$ for all $\lambda > 0$, $A \subset \mathbb{R}^n$.

(iii) $H^s(L(A)) = H^s(A)$ for any affine isometry $L : \mathbb{R}^n \to \mathbb{R}^n$, $A \subset \mathbb{R}^n$.

证明. (i) $\alpha(0) = 1$, therefore $H^0(\{x\}) = 1$ for any $x \in \mathbb{R}^n$. Since $\{x\}$ is H^s-measurable, $H^0(\{x_1, \cdots, x_i\}) = i$.

(ii) Trivial.

(iii) Trivial. □

定义 1.12.3. *A Borel regular measure μ on a metric space X is called uniformly distributed if*

$$0 < \mu(B(x, r)) = \mu(B(y, r)) < \infty, \quad \forall x, y \in X, 0 < r < \infty.$$

定理 1.12.4. *Let μ and ν be uniformly distributed Radon measures on $\mathbb{R}^n$. Then there exists a constant C such that $\mu = c\nu$.*

证明. Since μ and ν are Radon measures, it suffices to show that $\mu(U) = c\nu(U)$ for all open sets U. Since μ and ν are uniformly distributed, for $r > 0$ define

$$g(r) := \mu(B(x, r)), \quad h(r) := \nu(B(x, r))$$

Clearly,

$$\lim_{r\to 0} \frac{\nu(U \cap B(x, r))}{h(r)} = 1, \quad \forall x \in U$$

By Fatou's lemma,

$$\mu(U) = \int_U \lim_{r\to 0} \frac{\nu(U \cap B(x, r))}{h(r)} \mathrm{d}\mu(x) \tag{1.6}$$

$$\leq \liminf_{r\to 0} \frac{1}{h(r)} \int_U \nu(U \cap B(x, r)) \mathrm{d}\mu(x) \tag{1.7}$$

$$= \liminf_{r\to 0} \frac{1}{h(r)} \int_U \mu(U \cap B(y, r)) \mathrm{d}\nu(y) \tag{1.8}$$

$$\leq \nu(U) \liminf_{r\to 0} \frac{g(r)}{h(r)} \tag{1.9}$$

Similarly,

$$\nu(U) \leq \mu(U) \liminf_{r\to 0} \frac{h(r)}{g(r)}$$

From the last two estimates,

$$\lim_{r\to 0} \frac{g(r)}{h(r)} = \frac{\mu(U)}{\nu(U)}, \forall \text{ open } U$$

Denote the left-hand side by C, and the proof is complete. □

1.12.2 Hausdorff Dimension

定理 1.12.5. *Let $0 \le s < t < \infty$ and $A \subset \mathbb{R}^n$.*

(i) If $H^s(A) < \infty$, then $H^t(A) = 0$.

(ii) If $H^t(A) > 0$, then $H^s(A) = \infty$.

证明. (i) By definition, there exists $\delta > 0$ and a δ-covering $\{C_j\}$ such that

$$\sum_j \operatorname{diam}(C_j)^s \le H^s_\delta(A) \le H^s(A) + 1$$

As a result,

$$H^t_\delta(A) \le \sum_j \operatorname{diam}(C_j)^t \tag{1.10}$$

$$\le \delta^{t-s} \sum_j (\operatorname{diam} C_j)^s \tag{1.11}$$

$$\le \delta^{t-s}(H^s(A) + 1) \tag{1.12}$$

Letting $\delta \to 0$, we have $H^t(A) = 0$.

(ii) Is equivalent to (i). □

定义 1.12.4. *The Hausdorff dimension of a set $A \subset \mathbb{R}^n$ is*

$$\dim A := \sup\{s : H^s(A) > 0\} \tag{1.13}$$

$$:= \sup\{s : H^s(A) = \infty\} \tag{1.14}$$

$$:= \inf\{t : H^t(A) = 0\} \tag{1.15}$$

$$:= \inf\{t : H^t(A) < \infty\} \tag{1.16}$$

By Theorem 1.12.5, this is well-defined.

注 1.12.3. *(i) $\dim \mathbb{R}^n = n$. This follows from the observations $H^n(\mathbb{R}^n) > 0$ and $s > n \Rightarrow H^s(\mathbb{R}^n) = 0$.*

命题 1.12.1. *Let $A \subset \mathbb{R}^n, 0 \le s < \infty, 0 < \delta \le \infty$. The following are equivalent:*

(i) $H^s(A) = 0$

(ii) $H^s_\delta(A) = 0$

(iii) For all $\epsilon > 0$, there exist $E_1, E_2, \cdots \subset \mathbb{R}^n$ such that $A \subset \bigcup E_j$ and $\sum [\operatorname{diam}(E_j)]^s < \epsilon$.

证明. When $s = 0$, all are equivalent to $A = \varnothing$. Assume $s > 0$.

(i)⇒(ii) $0 \le H^s_\delta(A) \le H^s(A) \le 0$.

(ii)⇒(iii) By definition.

(iii)⇒(i) Let $\delta = \epsilon^{1/s}$. Then $H^s_\delta(A) < \delta^s$. Letting $\delta \to 0$, we obtain (i). □

命题 1.12.2. *(i) If $A \subset B \subset \mathbb{R}^n$, then $\dim A \le \dim B$.*

(ii) If $A_i \subset \mathbb{R}^n, i = 1, 2, \ldots$, then $\dim\left(\bigcup_{i=1}^\infty A_i\right) = \sup_i \dim A_i$.

证明. (i) Follows from $H^s(A) \le H^s(B)$ for all $s \ge 0$.

(ii) By (i), $\sup_i \dim A_i \le \dim\left(\bigcup_i A_i\right)$. Denote the RHS by t, then for all $\epsilon > 0$,

$$\infty = H^{t-\epsilon} \le \sum_i H^{t-\epsilon}(A_i)$$

In particular, there exists i such that $H^{t-\epsilon}(A_i) > 0$, which implies $\dim A_i \ge t - \epsilon$. As ϵ is arbitrary, we obtain the converse. □

定理 1.12.6. *Let $A \subset \mathbb{R}^n$, then*

$$\dim A \le \underline{\dim}_M(A)$$

证明. It suffices to show that for any $\beta > \underline{\dim}_M(A)$, we have $\dim A \le \beta$. Let $\alpha \in (\underline{\dim}_M A, \beta)$. By definition, for any $\delta_0 > 0$, there exists $\delta \in (0, \delta_0)$ and a δ-covering $\{E_j\}_{j=1}^J$ such that $J \le \delta^{-\alpha}$, so

$$\sum_j (\operatorname{diam} E_j)^\beta \le \delta^{\beta-\alpha} \le \delta_0^{\beta-\alpha}$$

Since δ_0 is arbitrary, by Proposition 1.12.1(iii), $H^\beta(A) = 0$, i.e., $\dim A \le \beta$. □

定理 1.12.7. *If $f : \mathbb{R}^m \to \mathbb{R}^n$ is Lipschitz, $0 \le s \le m$, and $A \subset \mathbb{R}^m$, then*

$$H^s(fA) \le \operatorname{Lip}(f)^s H^s(A) \tag{1.17}$$

In particular,

$$\dim(fA) \le \dim(A) \tag{1.18}$$

证明. Let $\{E_j\}$ be a δ-covering of A. Then $\{fE_j\}$ is a $\operatorname{Lip}(f)\delta$-covering of fA. In particular,

$$H^s_{\operatorname{Lip}(f)\delta}(fA) \le \operatorname{Lip}(f)^s H^s_\delta(A)$$

Letting $\delta \to 0$, we obtain (1.17) and (1.18). □

1.13　讲义: 几何测度论/ Chapter 2 Orthogonal Projection

原网页

The main references of this chapter are [Folland; Real Analysis, Chapter 4 and Chapter 7], [Evans, Gariepy; Measure theory and fine properties of functions, Chapter 1], [Mattila; Geometry of Sets and Measures in Euclidean Spaces-Fractals and Rectifiability, Chapter 1 and 8], and [Hochman; Lectures on fractal geometry and dynamics, Chapter 7]

1.14　Projections of Discrete Sets

Given $e \in S^1 := \{x \in \mathbb{R}^2 : |x| = 1\}$, we define

$$\pi_e(x) := (x \cdot e)e$$

which is the orthogonal projection of x to

$$l_e := \{te : t \in \mathbb{R}\}$$

Let $P \subset \mathbb{R}^2$ be a finite set such that $|P| = N$. We want to estimate the size of the exceptional set

$$E_P := \left\{ e \in S^1 : |\pi_e(P)| \leq \frac{N}{8} \right\}$$

引理 1.14.1. $|E_P| \leq N^2$.

证明. We may assume that directions in E_P are not parallel. Fixing $e \in E_P$, there exist two points $p_1, p_2 \in P$ such that

$$\pi_e(p_1) = \pi_e(p_2)$$

then the line through p_1 and p_2 is perpendicular to l_e. Since $l_e \neq l_{e'}$ when $e \neq e'$, we obtain

$$\begin{aligned} |E_P| &= |\{l_e : e \in E_P\}| \\ &\leq |\{(p_1, p_2) \in P^2 : p_1 \neq p_2\}| \\ &\leq N^2. \end{aligned}$$

□

定理 1.14.1. $|E_P| \leq 64N$.

注 1.14.1. *Theorem 1.14.1 is sharp in that there exists P such that $|E_P| \geq C \cdot N$ for some $C > 0$.*

命题 1.14.1 (Weak Szemerédi-Trotter)**.** *Let $P \subset \mathbb{R}^2$ be finite and $\mathscr{L}$ a finite collection of lines in $\mathbb{R}^2$. Then*

$$|\mathscr{I}(P, \mathscr{L})| \leq 4 \min\{|P||\mathscr{L}|^{1/2} + |\mathscr{L}|, |P|^{1/2}|\mathscr{L}| + |P|\}$$

where

$$\mathscr{I}(P, \mathscr{L}) := \{(p, l) \in P \times \mathscr{L} : p \in l\}.$$

注 1.14.2. *(Classic) Szemerédi-Trotter says*

$$|\mathscr{I}(P, \mathscr{L})| \lesssim |P|^{2/3}|\mathscr{L}|^{2/3} + |P| + |\mathscr{L}|.$$

Proof of Proposition 1.14.1. We prove only

$$|\mathscr{I}(P, \mathscr{L})| \leq 4(|P||\mathscr{L}|^{1/2} + |\mathscr{L}|).$$

Note that $|\mathscr{I}(P, \mathscr{L})| = \sum_{l \in \mathscr{L}} |P \cap l|$. By definition, we apply Cauchy-Schwarz to obtain

$$|\mathscr{I}(P, \mathscr{L})| \leq |\mathscr{L}|^{1/2} \left(\sum_{l \in \mathscr{L}} |P \cap l|^2 \right)^{1/2}.$$

By Fubini's theorem,

$$\begin{aligned} \sum_{l \in \mathscr{L}} |P \cap l|^2 &= \sum_{l \in \mathscr{L}} \sum_{p \in P} \sum_{q \in P} \chi_{p \in l} \chi_{q \in l} \\ &= \sum_{p \in P} \sum_{q \in P} \sum_{l \in \mathscr{L}} \chi_{p \in l} \chi_{q \in l} \end{aligned}$$

$$= \sum_{p\in P}\sum_{l\in\mathscr{L}} \chi_{p\in l} + \sum_{p\neq q}\sum_{l\in\mathscr{L}} \chi_{p\in l}\chi_{q\in l}$$
$$= |\mathscr{I}(P,\mathscr{L})| + |P|^2.$$

In summary,

$$|\mathscr{I}(P,\mathscr{L})| \leq |\mathscr{L}|^{1/2}(|\mathscr{I}(P,\mathscr{L})|^{1/2} + |P|).$$

If the first term dominates, i.e., $|\mathscr{I}|^{1/2} \geq |P|$, then

$$|\mathscr{I}(P,\mathscr{L})| \leq 4|\mathscr{L}|.$$

Otherwise,

$$|\mathscr{I}(P,\mathscr{L})| \leq |P||\mathscr{L}|^{1/2}.$$ □

Proof of Theorem 1.14.1. Fixing $e \in E_P$, define

$$\mathscr{L}_e := \{\pi_e^{-1}(t) : t \in \pi_e(P)\}$$

the family of lines perpendicular to e and intersecting P. Obviously, $|\mathscr{I}(P,\mathscr{L}_e)| = N$ as each point in P is contained in exactly one line in $\mathscr{L}_e$.

Define

$$\mathscr{L} := \bigcup_{e\in E_P} \mathscr{L}_e$$

then $|\mathscr{L}| \leq |E_P|\frac{N}{8}$. By Proposition 1.14.1,

$$\begin{aligned} |E_P|N &= |\mathscr{I}(P,\mathscr{L})| \\ &\leq 4(|P||\mathscr{L}|^{1/2} + |\mathscr{L}|) \\ &\leq 4\left(N\left(\frac{|E_P|N}{8}\right)^{1/2} + \frac{|E_P|N}{8}\right). \end{aligned}$$

Therefore $\left(\frac{1}{2}|E_P|N\right)^2 \leq 2|E_P|N^3$, that is,

$$|E_P| \leq 8N.$$ □

定理 1.14.2. *Let*

$$P := \{(m,n) \in \mathbb{Z}^2 : 0 \leq m,n \leq N\}$$

Let

$$S(P) := \left\{e_{mn} := \frac{(m,n)}{|(m,n)|} \in S^1 : 0 \leq m,n \leq \frac{N}{100}\right\}$$

be a subset of directions generated by P. Then

$$\pi_e(P) \leq \frac{N^2}{10} \quad \forall e \in S(P), \tag{1.19}$$
$$|S(P)| \gtrsim N^2. \tag{1.20}$$

In particular,

$$|E(P)| \geq |S(P)| \gtrsim N^2. \tag{1.21}$$

证明. Fix $e = e_{mn} = \frac{(m,n)}{|(m,n)|}$. Let $a_j = (m_j, n_j), j = 1, 2, \dots$ in P. If $\pi_e(a_1) \neq \pi_e(a_2)$, then $m_1 m + n_1 n \neq m_2 m + n_2 n$, which implies that

$$\begin{aligned} 0 &\neq |\pi_e(a_1) - \pi_e(a_2)| \\ &= \frac{|mm_1 + nn_1 - (mm_2 + nn_2)|}{|(m,n)|} \\ &\geq \frac{50}{N}. \end{aligned}$$

As a result, $\pi_e(P)$ is a $\frac{50}{N}$-separated subset of a segment of length $\leq 2N$. In particular,

$$|\pi_e(P)| \leq \frac{N^2}{25}$$

which verifies (1.19). We may assume $n \leq m$ by symmetry. Then $S(P)$ is the set of irreducible fractions $\frac{n}{m}$ with $n \leq m \leq \frac{N}{100}$, which is

$$\sum_{1 \leq k \leq \frac{N}{100}} \varphi(k) \tag{1.22}$$

with $\varphi(k) := |\{1 \leq m \leq k : \gcd(m, k) = 1\}|$. It is known

$$\sum_{1 \leq k \leq \frac{N}{100}} \varphi(k) \sim N^2$$

i.e., (1.20) holds. By (1.19),

$$S(P) \subset E_P$$

therefore (1.21) follows from (1.20). □

注 **1.14.3.** *A set A is δ-separated if $|a_1 - a_2| \geq \delta, \forall a_1 \neq a_2 \in A$.*

1.15 Weak Convergence of Measures

定理 1.15.1 (Urysohn's Lemma)**.** *If X is an LCH space, and $K \subset U \subset X$, where K is compact, and U is open. Then $\exists f \in C(X, [0, 1])$ such that*

$$K \prec f \prec U.$$

Let $C_c(X)$ be the space of continuous functions on X with compact support. For $f \in C_c(X)$, define

$$\|f\|_u := \sup_{x \in X} |f(x)|.$$

A linear functional I on $C_c(X)$ is called positive if $I(f) \geq 0$ whenever $f \geq 0$.

命题 1.15.1. *Let I be a positive linear functional on $C_c(X)$. Then, $\forall$ compact $K \subset X$, $\exists$ a constant C_k s.t.*

$$|I(f)| \leq C_k \|f\|_u$$

$\forall f \in C_c(X)$ with supp $f \subset K$.

证明. We may assume that f is real-valued. Given K, $\exists \phi \in C_c(X,[0,1])$ s.t. $\phi = 1$ on K. Suppose supp $f \subset K$, then

$$|f| \leq \|f\|_u \phi$$

i.e., $\phi\|f\|_u - f, \|f\|_u\phi + f \geq 0$. As a result,

$$I(\phi\|f\|_u - f),\ I(\|f\|_u\phi + f) \geq 0$$

i.e., $|I(f)| \leq I(\phi)\|f\|_u$. □

命题 1.15.2 (Riesz Representation)**.** *If I is a positive linear functional on $C_c(\mathbb{R}^n)$, then $\exists$ a unique Radon measure μ on $\mathbb{R}^n$ s.t.*

$$I(f) = \int f \mathrm{d}\mu, \quad \forall f \in C_c(\mathbb{R}^n).$$

Moreover, μ satisfies

(i) $\mu(U) = \sup\{I(f) : f \in C_c(\mathbb{R}^n), f \prec U, U \text{ open}\}$,

(ii) $\mu(K) = \inf\{I(f) : f \in C_c(\mathbb{R}^n), \chi_K \leq f, K \text{ compact}\}$.

定义 1.15.1. *Let $\mu, \mu_k (k = 1, 2, \dots)$ be Radon measures on $\mathbb{R}^n$. If*

$$\lim_{k\to\infty} \int_{\mathbb{R}^n} f \mathrm{d}\mu_k = \int_{\mathbb{R}^n} f \mathrm{d}\mu, \quad \forall f \in C_c(\mathbb{R}^n),$$

then we say the measures $\{\mu_k\}_{k=1}^{\infty}$ converge weakly to μ, written

$$\mu_k \to \mu.$$

例 1.15.1. *Let δ_x be the Dirac measure at x, which is defined by*

$$\delta_x(A) = \begin{cases} 1 & x \in A \\ 0 & x \notin A \end{cases}$$

Then $\delta_i \to 0$ as $i \to \infty$.

例 1.15.2. *Let $\mu_k = \frac{1}{k}\sum_{i=1}^{k} \delta_{\frac{i}{k}}$, then $\mu_k \to \mathcal{L}^1|_{[0,1]}$ as $k \to \infty$.*

引理 1.15.1. *$C_c(\mathbb{R}^n)$ is separable under the norm $\|\cdot\|_u$.*

定理 1.15.2. *Let $\{\mu_k\}_{k=1}^{\infty}$ be a sequence of Radon measures on $\mathbb{R}^n$ satisfying*

$$\sup_k \mu_k(K) < \infty \quad \forall \text{ compact } K \subset \mathbb{R}^n \tag{1}$$

Then $\exists$ a subsequence $\{\mu_{k_j}\}_{j=1}^{\infty}$ and a Radon measure μ s.t.

$$\mu_k \to \mu.$$

证明. We consider first the case

$$\sup_k \mu_k(\mathbb{R}^n) < \infty. \tag{2}$$

Step 1: By Lemma 1.15.1, $\exists$ a countable dense subset $\{f_k\}_{k=1}^{\infty}$ of $C_c(\mathbb{R}^n)$. As (2) implies that $\left\{\int f_1 \mathrm{d}\mu_j\right\}_j$ is bounded, we can find a subsequence $\{\mu_j^1\}_{j=1}^{\infty}$ and $a_1 \in \mathbb{R}$ s.t.

$$\int f_1 \mathrm{d}\mu_j^1 \to a_1$$

Repeating this argument, we can find a subsequence $\{\mu_j^k\}_j$ of $\{\mu_j^{k-1}\}_j$ s.t.

$$\int f_k \mathrm{d}\mu_j^k \to a_k \tag{3}$$

Setting $v_j := \mu_j^j$, we obtain from (3) that, $\forall k$,

$$\int f_k \mathrm{d}\mu_j \to a_k \quad \text{as} \quad j \to \infty.$$

Step 2: Define a linear functional L on span$(\{f_k\})$ by $L(f_k) = a_k$. Then

$$|L(f_k)| \le \|f_k\|_u M,$$

where $M := \sup_j \mu_j(\mathbb{R}^n) < \infty$. As a result, we can extend L uniquely to a bounded linear functional $\bar{L}$ on $C_c(\mathbb{R}^n)$. Moreover, $\bar{L}$ is positive. Actually, $\forall f \ge 0$ in $C_c(\mathbb{R}^n)$, $\exists f_k$ s.t. $\|f_k - f\|_u < \frac{\epsilon}{M}$. Then

$$\bar{L}(f) = \bar{L}(f - f_k) + L(f_k) \ge -\epsilon - \epsilon$$

Applying Proposition 1.15.2, $\exists$ a Radon measure μ on $\mathbb{R}^n$ s.t.

$$\bar{L}(f) = \int_{\mathbb{R}^n} f \mathrm{d}\mu, \quad \forall f \in C_c(\mathbb{R}^n).$$

Step 3: Fix $f \in C_c(\mathbb{R}^n)$. $\forall \epsilon > 0$, $\exists f_k$ s.t.

$$\|f - f_k\|_u < \frac{\epsilon}{4M}$$

We choose J large such that $\forall j > J$,

$$\left|\int f_k \mathrm{d}\nu_j - \int f_k d\mu\right| < \frac{\epsilon}{2}$$

Then $\forall j > J$, we have

$$\left|\int f \mathrm{d}\nu_j - \int f \mathrm{d}\mu\right| \le \left|\int (f - f_k) d\nu_j\right| + \left|\int f_k \mathrm{d}\nu_j - \int f_k \mathrm{d}\mu\right| + \left|\int (f_k - f) \mathrm{d}\mu\right| < \epsilon.$$

Step 4: We discuss below the case (2) fails, when we still have (1). Denoting $\mu_{k,l} := \mu_k|_{B(l)}$, where $B(l) := \{x \in \mathbb{R}^n : |x| \le l\}$. By previous arguments, we can find $\{\mu_k^j\} \subset \{\mu_k^{j-1}\}$ s.t.

$$\mu_{k,j}^j \to \sigma_j$$

Letting $\nu_k := \mu_k^k$, we have

$$\nu_{k,l} \to \sigma_l, \quad \forall l.$$

$\forall f \in C_c(\mathbb{R}^n)$ with supp $f \subset B(l)$, when $l', l'' \geq l$, we have

$$\begin{aligned}\int f \mathrm{d}\sigma_{l'} &= \lim_{k\to\infty} \int f \mathrm{d}\nu_{k,l'} \\ &= \lim_{k\to\infty} \int f \mathrm{d}\nu_k \\ &= \lim_{k\to\infty} \int f \mathrm{d}\nu_{k,l''} \\ &= \int f \mathrm{d}\sigma_{l''}.\end{aligned}$$

Therefore, we can define

$$L(f) := \lim_{k\to\infty} \int f \mathrm{d}\nu_k$$

Applying Proposition 1.15.2, $\exists$ a Radon measure μ such that $L(f) = \int f \mathrm{d}\mu$, i.e., $\nu_k \to \mu$. □

定义 1.15.2. *Let f and g be real-valued, μ a Radon measure. The convolution of f and g is defined by*

$$(f * g)(x) := \int f(x-y)g(y)\mathrm{d}y \tag{1.23}$$

$$(f * \mu)(x) := \int f(x-y)\mathrm{d}\mu(y) \tag{1.24}$$

$$\mu * \nu(A) := \int\int 1_A(x+y)\mathrm{d}\mu(x)\mathrm{d}\nu(y). \tag{1.25}$$

($\mathrm{d}\mu = g\mathrm{d}\mathcal{L}^n$)

注 1.15.1. $\mathrm{d}\mu(x) = g(x)\mathrm{d}x \Rightarrow f * \mu = f * g$.

定义 1.15.3. *Let $\{\psi_\epsilon\}_{\epsilon>0}$ be a family of non-negative continuous functions on $\mathbb{R}^n$. If $\forall \epsilon > 0$,*

$$\text{supp}\, \psi_\epsilon \subset B(0,\epsilon),$$

and

$$\int \psi_\epsilon \,\mathrm{d}x = 1,$$

then we call $\{\psi_\epsilon\}$ an approximate identity.

例 1.15.3. *Let $\psi \in C_c^\infty(\mathbb{R}^n)$ with* supp $\psi \subset B(0,1)$ *and $\int \psi = 1$. Let $\psi_\epsilon(x) := \epsilon^{-n}\psi(\epsilon^{-1}x)$. Then $\{\psi_\epsilon\}$ is an approximate identity.*

定理 1.15.3. *Let $\{\psi_\epsilon\}$ be an A.I., and μ a Radon measure on $\mathbb{R}^n$. Then*

$$\psi_\epsilon * \mu \in C^\infty(\mathbb{R}^n) \tag{1.26}$$

and

$$\psi_\epsilon * \mu \to \mu \quad \text{as } \epsilon \to 0,$$

i.e.

$$\int f(x)(\psi_\epsilon * \mu)(x)\,\mathrm{d}x \to \int f(x)\,\mathrm{d}\mu(x) \quad \forall f \in C_c(\mathbb{R}^n). \tag{1.27}$$

If $\mu(\mathbb{R}^n) < \infty$, then (1.27) *also holds for all uniformly continuous bounded functions.*

证明. Equation (1.26) follows from dominated convergence and smoothness of the mollifier. To prove Equation (1.27), observe that

$$\begin{aligned}
&\int f(x)(\psi_\epsilon * \mu)(x)\,\mathrm{d}x - \int f(y)\,\mathrm{d}\mu(y) \\
&= \int \left[\int f(x)\psi_\epsilon(x-y)\,\mathrm{d}x - \int f(y)\psi_\epsilon(x)\,\mathrm{d}x \right] \mathrm{d}\mu(y) \\
&= \int \left[\int f(x+y)\psi_\epsilon(x)\,\mathrm{d}x - \int f(y)\psi_\epsilon(x)\,\mathrm{d}x \right] \mathrm{d}\mu(y) \\
&= \int \int (f(x+y)-f(y))\psi_\epsilon(x)\,\mathrm{d}x\,\mathrm{d}\mu(y).
\end{aligned}$$

For ϵ small and f uniformly continuous on a neighbourhood of supp f, the inner difference is small; hence

$$\left| \int f(x)(\psi_\epsilon * \mu)(x)\,\mathrm{d}x - \int f\,\mathrm{d}\mu \right| \le \mu(\mathrm{supp}(f) + B(0,\epsilon_0))\delta,$$

which tends to 0 as $\epsilon \to 0$. □

定理 1.15.4. *Let $\mu, \mu_k (k = 1, 2, \dots)$ be Radon measures on $\mathbb{R}^n$. The following are equivalent:*

(i) $\mu_k \to \mu$*;*

(ii) $\limsup_{k\to\infty} \mu_k(K) \le \mu(K)$ *for all compact $K \subset \mathbb{R}^n$, and $\mu(U) \le \liminf_{k\to\infty} \mu_k(U)$ for all open $U \subset \mathbb{R}^n$;*

(iii) $\lim_{k\to\infty} \mu_k(B) = \mu(B)$ *for every bounded Borel set B with $\mu(\partial B) = 0$.*

证明. (i) ⇒ (ii): Fix compact K and open U. Choose $f \in C_c(\mathbb{R}^n)$ with $K \prec f \prec U$ (Urysohn). Then

$$\mu(K) \le \int f\,\mathrm{d}\mu = \lim_{k\to\infty} \int f\,\mathrm{d}\mu_k \le \liminf_{k\to\infty} \mu_k(U),$$

and since μ is Radon we get $\mu(U) \le \liminf_{k\to\infty} \mu_k(U)$. The other inclusion is similar.

(ii) ⇒ (iii): Let B° denote the interior of B. By assumption $\mu(B^\circ) = \mu(B) = \mu(\overline{B})$; hence

$$\begin{aligned}
\mu(B) &= \mu(B^\circ) \\
&\le \liminf_{k\to\infty} \mu_k(B^\circ) \\
&\le \limsup_{k\to\infty} \mu_k(\overline{B}) \\
&= \mu(\overline{B}) \\
&= \mu(B).
\end{aligned}$$

(iii) ⇒ (i): Standard approximation of $f \in C_c$ by simple functions. The proof follows by slicing the range of f into small intervals and using sets with boundary of measure zero to pass to the limit (see text for details). □

1.16 Frostman's Lemma

定义 1.16.1. *Let μ be a Borel measure on $\mathbb{R}^n$. The support of μ, denoted* supp μ*, is the smallest closed set F with*

$$\mu(F^c) = 0.$$

Equivalently,

$$\operatorname{supp}\mu := \Big(\bigsqcup\{V \subset \mathbb{R}^n : V \text{ open},\ \mu(V) = 0\}\Big)^c.$$

注 1.16.1. *(i) If $f \in C_c(\mathbb{R}^n)$, $f \geq 0$, and $\mathrm{d}\mu_f := f\,\mathrm{d}\mathcal{L}^n$ then $\operatorname{supp} f = \operatorname{supp}\mu_f$.*

(ii) If $\operatorname{supp} g \subset (\operatorname{supp}\mu)^c$ then $\int g\,\mathrm{d}\mu = 0$.

(iii) $\operatorname{supp}\mu = \{x : \forall r > 0,\ \mu(B(x,r)) > 0\}$.

(iv) Example: enumerating rationals $Q = \{q_1, q_2, \dots\}$ and setting $\mu = \sum 2^{-i}\delta_{q_i}$ yields $\operatorname{supp}\mu = \mathbb{R}$ while $\mu(\mathbb{R} \setminus \mathbb{Q}) = 0$.

定理 1.16.1 (Mass distribution principle)**.** *If E supports a strictly positive Borel measure μ satisfying*

$$\mu(B(x,r)) \leq Cr^s, \quad \forall x \in \mathbb{R}^n,\ r > 0,$$

where $s, C \in (0,\infty)$, then

$$\mathcal{H}^s(E) \geq \mathcal{H}^s_\infty(E) \geq \frac{\mu(E)}{C}.$$

In particular, $\dim E \geq s$.

证明. Let $\{U_i\}$ be a cover of E and denote $r_i := \operatorname{diam}(U_i)$. Choose $x_i \in U_i$ so $U_i \subset B(x_i, r_i)$. Then

$$\mu(U_i) \leq \mu(B(x_i, r_i)) \leq Cr_i^s.$$

Summing yields

$$\sum r_i^s \geq \frac{1}{C}\sum \mu(U_i) \geq \frac{\mu(E)}{C}.$$

Taking infimum over covers proves the claim. □

定义 1.16.2. *For $A \subset \mathbb{R}^n$ let*

$$M(A) := \{\mu : \mu \text{ Radon, compactly supported},\ \operatorname{supp}\mu \subset A,\ 0 < \mu(\mathbb{R}^n) < \infty\}.$$

定义 1.16.3. *An s-Frostman measure on A is $\mu \in M(A)$ such that for some $C > 0$,*

$$\mu(B(x,r)) \leq Cr^s, \quad \forall x, r > 0.$$

定义 1.16.4. *A dyadic cube of scale 2^{-m} is*

$$\{x :\ k_i 2^{-m} \leq x_i < (k_i+1)2^{-m},\ i = 1, \dots, n\}.$$

Denote by $\mathscr{D}_m$ the family of dyadic cubes of side length 2^{-m}.

定理 1.16.2 (Frostman's Lemma)**.** *Let B be Borel in $\mathbb{R}^n$. Then $\mathcal{H}^s(B) > 0$ iff there exists an s-Frostman measure supported on B. Moreover $\mu(B) \geq c\,\mathcal{H}^s_\infty(B)$ for some $c = c(n) > 0$.*

证明. **AI 在此处进行了不当的简化。待完善**

We sketch the usual construction for compact B. Fix $Q_0 \in \mathscr{D}_{m_0}$ with $B \subset Q_0$. For large $m \geq m_0$ define a preliminary measure μ_m^m by putting on each dyadic cube $Q \in \mathscr{D}_m$ meeting B the mass 2^{-ms} distributed uniformly in Q, and 0 otherwise. Iteratively renormalize on coarser dyadic grids so that no dyadic cube of side 2^{-k} carries mass exceeding 2^{-ks}. Passing to a weak limit along $m \to \infty$ yields an s-Frostman measure; the estimates give $\mu(B) \gtrsim \mathcal{H}^s_\infty(B)$. □

1.17　讲义: 几何测度论/ Chapter 3 Differentiation of Measures

原网页

The main references of this chapter are [Evans, Gariepy; Measure theory and fine properties of functions, Chapter 1], [Mattila; Geometry of Sets and Measures in Euclidean Spaces-Fractals and Rectifiability, Chapter 2].

$$f \in L^1_{\text{loc}}(\mathbb{R}^n) \Rightarrow \lim_{r\to 0} \frac{\int_{B(x,r)} f}{|B(x,r)|} = f \quad \text{a.e.}$$

1.18　Vitali's covering Lemma

定义 1.18.1. $B(x,r) := \{y \in \mathbb{R}^n : |x-y| \le r\}$ *and* $5B(x,r) := B(x,5r)$. *A family*

$$\mathscr{F} = \{B(x_\alpha, r_\alpha) : \alpha \in \mathcal{A}\}$$

is a covering of $A \subset \mathbb{R}^n$ *if* $A \subset \bigcup_{\alpha\in\mathcal{A}} B(x_\alpha, r_\alpha)$. *The covering is fine if for every* $x \in A$ *we have*

$$\inf\{r > 0 :\ x \in B(y,r) \in \mathscr{F}\} = 0.$$

定理 1.18.1. *Let* $\mathscr{F}$ *be a collection of closed balls in* $\mathbb{R}^n$ *satisfying* $\sup_\alpha r_\alpha < \infty$. *Then there exists a countable subcollection* $\mathscr{G}$ *of disjoint balls in* $\mathscr{F}$ *with*

$$\bigcup_{B\in\mathscr{F}} B \subset \bigcup_{B\in\mathscr{G}} 5B.$$

证明. Let $R_i := 2^{-i}\sup_\alpha r_\alpha, i \ge 0$, $F_i := \{B \in \mathscr{F} : R_i < r(B) \le R_{i-1}\}$.

A sequence $\mathscr{G}_n$ with $\mathscr{G}_n \subset \mathscr{F}_n$ is defined inductively as follows:

First let $\mathscr{G}_1$ be the maximal subcollection of disjoint balls in $\mathscr{F}_1$. Assuming that $\mathscr{G}_1, \dots, \mathscr{G}_{k-1}$ have been selected, let

$$\mathscr{H}_k = \{B \in \mathscr{F}_k : B \cap C = \varnothing, \forall C \in \mathscr{G}_1 \cup \mathscr{G}_1 \cup \cdots \cup \mathscr{G}_{k-1}\},$$

and let $\mathscr{G}_k$ be a maximal subcollection of disjoint balls in $\mathscr{H}_k$. We obtain a subcollection

$$\mathscr{G} := \bigcup_{n=0}^{\infty} \mathscr{G}_n$$

of $\mathscr{F}$.

Claim. $\forall B \in \mathscr{F}$, $\exists B' \in \mathscr{G}$ *with* $r(B') \ge \frac{r(B)}{2}$ *such that* $B \cap B' \ne \varnothing$.

This is clear by definition.

By our claim, $B \subset 5B'$, hence

$$\bigcup_{B\in\mathscr{F}} B \subset \bigcup_{B'\in\mathscr{G}} 5B'.$$

□

推论 1.18.1. *If $\mathscr{F}$ is a fine covering of A with $\sup r_\alpha < \infty$, then there exists a countable disjoint subcollection $\mathscr{G} \subset \mathscr{F}$ such that for each finite subcollection $\{B_1, \ldots, B_m\} \subset \mathscr{F}$*

$$A \setminus \bigcup_{k=1}^{m} B_k \subset \bigcup_{B \in \mathscr{G} \setminus \{B_1, \ldots, B_m\}} 5B. \tag{1.28}$$

证明. Take $\mathscr{G}$ as in Theorem 1.18.1. Then $\forall x \in A \setminus \bigcup_{k=1}^{m} B_k, \exists B \in \mathscr{F}$ such that $x \in B$ and $B \cap B_k = \varnothing$ as $\mathscr{F}$ is fine. Applying claim, $\exists B' \in \mathscr{G}$ such that $B \cap B' \neq \varnothing$. In particular, $B' \in \{B_1, \ldots, B_m\}$, which implies (1.28). □

定理 1.18.2. *Let $\mathscr{F}$ be a fine covering of a measurable set $A \subset \mathbb{R}^n$. Then there exists a countable disjoint subcollection $\mathscr{G}$ of balls in $\mathscr{F}$ such that*

$$\left| A \setminus \bigcup_{B \in \mathscr{G}} B \right| = 0.$$

Moreover, for every $\epsilon > 0$, we can find $\mathscr{G}$ such that

$$\sum_{B \in \mathscr{G}} |B| \leq |A| \, (1 + \varepsilon).$$

证明. We may assume that A is bounded by decomposing $\mathbb{R}^n = \bigcup_i \overline{Q_i}$. Take an open set $U \supset A$ such that $|U| \leq |A| \, (1 + \varepsilon)$. Applying Theorem 1.18.1, we can find disjoint balls $B_1, \ldots, B_k \in \mathscr{F}$ contained in U such that

$$6^{-n} \, |A| \leq \sum_{i=1}^{k_1} |B_i| \, .$$

Then $A_1 := A \setminus \bigcup_{i=1}^{k} B_i$ satisfies $|A_1| \leq (1 - 7^{-n}) \, |A|$ by taking ε small enough. Find $U_1 \supset A_1$ with $|U_1| \, |A_1| \, (1 + \varepsilon)$ then we can find disjoint balls $B_{k_1+1}, \ldots, B_{k_2} \in \mathscr{F}$ contained in U_1 such that

(1) $6^{-n} \, |A_1| \leq \sum_{i=k_1+1}^{k_2} |B_i|$;

(2) $B_{k_1+1}, \ldots, B_k$ are disjoint.

Now $A_2 := A \setminus \bigcup_{i=1}^{k_2} B_i$ satisfies $|A_2| \leq (1 - 7^{-n})^2 \, |A|$.

Repeating above process we obtain a countable subcollection $\mathscr{G} = \{B_i\}$ of disjoint balls in $\mathscr{F}$ satisfying

$$\left| A \setminus \bigcup_{i=1}^{\infty} B_i \right| \leq \left| A \setminus \bigcup_{i=1}^{k_j} B_i \right| \leq (1 - 7^{-n})^j |A| \to 0,$$

and

$$\sum_i |B_i| = \left| \bigcup_i B_i \right| \leq |U| \leq |A|(1 + \varepsilon).$$

□

推论 1.18.2. *Let $U \subset \mathbb{R}^n$ be open and $\delta > 0$. Then there exists a countable collection $\mathscr{G}$ of disjoint closed balls in U with $r(B) < \delta$ for all $B \in \mathscr{G}$ and*

$$|U \setminus \bigcup_{B \in \mathscr{G}} B| = 0.$$

注 1.18.1. 上面的命题性质很好, 但是都用到了一个事实: $|5B| = 5^n |B|$. 这对 *Lebesgue* 测度当然成立, 但是对一般的 *Radon* 测度并没有这件事. 很简单地, 考虑仅支撑在一个较低维数集合上的 *Radon* 测度. 对于一般的测度, 需要研究更精细的覆盖引理.

1.19 Besicovitch's covering lemma

定理 1.19.1. *There exist integers $P(n)$ and $Q(n)$ such that: if $\mathscr{F} = \{B_\alpha : \alpha \in \mathcal{A}\}$ is a family of balls in $\mathbb{R}^n$ with $M_1 = \sup r_\alpha < \infty$, and A is the set of centers, then*

(i) there exists a countable subcollection $\mathscr{G} \subset \mathscr{F}$ covering A with

$$\chi_A \le \sum_{B\in\mathscr{G}} \chi_B \le P(n);$$

(ii) there exist $Q(n)$ countable collections $\mathscr{G}_1, \dots, \mathscr{G}_{Q(n)}$ of disjoint balls in $\mathscr{F}$ such that

$$A \subset \bigcup_{i=1}^{Q(n)} \bigcup_{B\in\mathscr{G}_i} B.$$

引理 1.19.1. *Let $a, b \in \mathbb{R}^2$ satisfy $0 < |a| < |a-b|$ and $0 < |b| < |a-b|$. Then the angle between a and b is at least $\pi/3$, i.e.*

$$\left|\frac{a}{|a|} - \frac{b}{|b|}\right| \ge 1.$$

证明. We may assume $a = (|a|, 0)$. Then $0 < |a| < |a-b|$ implies $b \notin B(a, |a|)$, and $0 < |b-0| < |b-a|$ implies $b_1 < \frac{|a|}{2} (b = (b_1, b_2))$. The conclusion follows. □

引理 1.19.2. *There exists $N(n)$ with the following property. If points $a_1, \dots, a_k \in \mathbb{R}^n$ and radii $r_1, \dots, r_k > 0$ satisfy*

$$a_i \notin B(a_j, r_j)\ \forall j \ne i, \qquad \bigcap_{i=1}^{k} B(a_i, r_i) \ne \emptyset,$$

then $k \le N(n)$.

证明. Assume $0 \in \bigcap_{i=1}^k B(a_i, r_i)$. Applying Lemma 1.19.1, we see that $\left\{e_i = \frac{a_i}{|a_i|} : i = 1, \dots, k\right\}$ are 1-separated. And the lemma follows. □

Proof of Theorem 1.19.1. First we assume that A is bounded. Choose $x_1 \in A$ such that $r_{x_1} \ge \frac{M_1}{2}$. (By taking the largest ball with the same center x, for each $x \in A$ we determine a unique ball with radii r_x in $\mathscr{F}$ and thereby get a subfamily of $\mathscr{F}$) Inductively we choose

$$x_{l+1} \in A \setminus \bigcup_{i=1}^{l} B(x_i, r_{x_i})$$

satisfying $r_{l+1} \ge \frac{M_1}{2}$. Since A is bounded, and $\{x_i\}$ are $\frac{M_1}{2}$-separated, we stop in finitely many steps, say k_1. Let $M_2 := \sup_{x\in A\setminus\bigcup_{i=1}^k B(x_i, r_{x_i})} r_x$. Then $M_2 \le \frac{M_1}{2}$. Similarly, we choose $x_{k_1+1}, \dots, x_{k_2}$. Repeating this process, we obtain $x_1, x_2, \dots, x_{k_1}, \dots, x_{k_2}, \dots$. Denoting $I_j := \{k_{j-1}+1, \dots, k_j\}$, $B_i := B(x_i, r_{x_i})$, we have

$$\frac{M_j}{2} < r_{x_i} \le M_j, \forall i \in I_j, \tag{1.29}$$

$$x_{l+1} \in A \setminus \bigcup_{i=1}^{l} B_i \forall l, \tag{1.30}$$

and(by (1.29) and (1.30))

$$x_i \in A \setminus \bigcup_{m\ne j} \bigcup_{i'\in I_m} B_{i'} \quad \forall i \in I_j. \tag{1.31}$$

(i) $A \subset \bigcup_i B_i$ as $M_j \to 0$. Actually, for $x \in A$ with $r_x \in (M_{j+1}, M_j]$, we have $x \in \bigcup_{m\le j}\bigcup_{i\in I_m} B_i$ by the choice of I_m. Let $\mathscr{G} := \{B_i : i \ge 1\}$. If $x \in B_{i_m} \in \mathscr{G}$ for $m = 1, \dots, p$, then $p \le 16^n N(n) =: P(n)$. Actually i_m belongs to at most $N(n)$ many I_j by Lemma 1.19.2 and (1.31). In each I_j, the number of balls involved is $\le 16^n$ since $x_{i_m} \in (B, M_j)$ and $\{x_{i_m}\}_{i_m\in I_j}$ are $\frac{M_j}{2}$-separated.

(ii) We may assume $r(B_i) \ge r(B_{i+1})$. Let $B_{1,1} = B_1$ and choose inductively $B_{1,k+1}$ as the ball B_i with the smallest i such that

$$B_i \cap \bigcup_{l=1}^{k} B_{1,l} = \varnothing.$$

Then we denote the collection of selected balls by $\mathscr{G}_1$. Let $B_{2,1}$ be the ball B_i with the smallest i such that $B_i \notin \mathscr{G}_1$. Then we can define $\mathscr{G}_2, \mathscr{G}_3, \dots, \mathscr{G}_m$. If we can show that

$$A \setminus \bigcup_{k=1}^{m} \bigcup_{B\in\mathscr{G}_k} B \ne \varnothing \Longrightarrow m \le 4^n P(n) + 1,$$

we complete the proof. Suppose $x \in A \setminus \bigcup_{k=1}^m \bigcup_{B\in\mathscr{G}_k} B$, then $\exists B_i \in \mathscr{G} \setminus \bigcup_{k=1}^m \mathscr{G}_k$ such that $x \in B_i$. By the choice of $\mathscr{G}_k$, for all k, $\exists B_{k,i_k} \in \mathscr{G}_k$ with $B_i \cap B_{k,i_k} \ne \varnothing$ and $r_i \le r_{k,i_k}$. As $\{B_{k,i_k}\}$ are finitely overlapping by (i), we can find balls $B'_k \subset 2B_i \cap B_{k,i_k}$ with $\frac{r_i}{2}$ such that $\{B'_k\}$ are also finitely overlapping. Observing that

$$m\left(\frac{r_i}{2}\right)^n \le \int \sum_{k=1}^{m} \chi_{B'_k} \le P(n)(2r_i)^n,$$

we obtain $m \le 16^n P(n)$.

To remove the restraint that A is bounded, we define

$$A_l := \{x \in A : 10M_1(l-1) \le |x| \le 10M_1 l\}$$

and apply the conclusion to A_l. □

定理 1.19.2. *Let $\mathscr{F}$ be a covering of $A \subset \mathbb{R}^n$ satisfying*

$$\inf\{r : B(x,r) \in \mathscr{F}\} = 0 \quad \forall x \in A,$$

(记录者注: 注意条件和 fine covering 的区别) and μ a Radon measure. Then $\exists$ a countable subcollection $\mathscr{G}$ of disjoint balls in $\mathscr{F}$ such that

$$\mu\left(A \setminus \bigcup_{B\in\mathscr{G}} B\right) = 0.$$

证明. We may assume that A is bounded. Take an open set $U \supset A$ such that $\mu(U) \le \mu(A)(1+\epsilon)$. Applying Theorem 1.19.1, we can find disjoint balls $B_1, \dots, B_m \subset U$ such that $\mu(A) \le 2Q(n)\sum_{i=1}^m \mu(B_i)$, which implies that

$$\mu\left(A \setminus \bigcup_{i=1}^{m} B_i\right) \le u\mu(A)$$

for $u = 1 - \frac{1}{10Q(n)} < 1$.

然后重复 Vitali 覆盖引理的证明过程, 注意到 u 与 B_i 的选取无关. 在每一步选取与上一步 B_i 不相交的开集挖去 (具体过程待补充). □

1.20 Differentiation of measures

定义 1.20.1. *Let μ and λ be locally finite Borel measures on $\mathbb{R}^n$. Define the upper and lower differentiations of μ with respect to λ at x:*

$$\overline{D}(\mu,\lambda,x) := \limsup_{r\to 0}\frac{\mu(B(x,r))}{\lambda(B(x,r))},$$
$$\underline{D}(\mu,\lambda,x) := \liminf_{r\to 0}\frac{\mu(B(x,r))}{\lambda(B(x,r))}.$$

At the point x where the limit exists we define

$$D(\mu,\lambda,x) = \lim_{r\to 0}\frac{\mu(B(x,r))}{\lambda(B(x,r))}.$$

定义 1.20.2. *μ is absolutely continuous with respect to λ if*

$$\lambda(A)=0 \Rightarrow \mu(A)=0 \quad \forall A\subset\mathbb{R}^n.$$

for which we write $\mu \ll \lambda$.

定理 1.20.1. *Let μ and λ be Radon measures on $\mathbb{R}^n$. Then*

(i) $D(\mu,\lambda,x)$ exists and is finite for λ-a.e. $x\in\mathbb{R}^n$;

(ii) For every Borel set $B\subset\mathbb{R}^n$, $\displaystyle\int_B D(\mu,\lambda,x)\,\mathrm{d}\lambda(x)\le\mu(B)$, with equality when $\mu\ll\lambda$;

(iii) $\mu\ll\lambda$ if and only if $\underline{D}(\mu,\lambda,x)<\infty$ for μ-a.e. x.

引理 1.20.1. *Let μ,λ be Radon measures, $0<t<\infty$, and $A\subset\mathbb{R}^n$. Then*

(i) If $\underline{D}(\mu,\lambda,x)\le t$ for all $x\in A$ then $\mu(A)\le t\lambda(A)$;

(ii) If $\overline{D}(\mu,\lambda,x)\ge t$ for all $x\in A$ then $\mu(A)\ge t\lambda(A)$.

证明. For (i) pick open $U\supset A$ with $\lambda(U)\le\lambda(A)+\varepsilon$. Applying Theorem 1.19.2, we obtain a countable collection of disjoint balls $B_i\subset U$ such that $\mu(B_i)\le(t+\varepsilon)\lambda(B_i)$, and $\mu\big(A\setminus\bigcup B_i\big)=0$. Then

$$\mu(A)\le(t+\varepsilon)\sum\lambda(B_i)\le(t+\varepsilon)(\lambda(A)+\varepsilon).$$

Let $\varepsilon\to 0$ to conclude. The proof of (ii) is similar. □

Proof of Theorem 1.20.1. (i) For any ball B, we define

$$B_{s,t} := \{x\in B : \underline{D}(\mu,\lambda,x)\le s<t\le\overline{D}(\mu,\lambda,x)\},$$

and

$$B_\mu := \{x\in B : \overline{D}(\mu,\lambda,x)\ge\mu\}.$$

Applying Lemma 1.20.1, for $s<t$,

$$t\lambda(B_{s,t})\le\mu(B_{s,t})\le s\lambda(B_{s,t}),$$

therefore $\lambda(B_{s,t})=0$, and $D(\mu,\lambda,x)$ exists for λ-a.e. x. Moreover,

$$\lim_{u\to\infty}\lambda(B_u)\le\lim_{u\to\infty}\frac{1}{u}\mu(B)=0.$$

Therefore $\overline{D}(\mu,\lambda,x)<\infty$ for λ-a.e. x.

(ii) Fix $t > 1$, and define

$$B^p := \{x \in B : D(\mu, \lambda, x) \in (t^p, t^{p+1}]\}.$$

As

$$\begin{aligned}
&\int_B D(\mu, \lambda, x)\, \mathrm{d}\lambda x \\
\leq &\sum_p \int_{B^p} D(\mu, \lambda, x)\, \mathrm{d}\lambda x \\
\leq &t \sum_p t^p \lambda(B^p) \\
\leq &t \sum_p \mu(B^p) \\
\leq &t\mu(B)
\end{aligned}$$

and let $t \to 1$. When $\mu \ll \lambda$, we apply (i) to show that

$$D(\mu, \lambda, x) \in (0, \infty) \quad \text{for} \quad \mu\text{-a.e. } x,$$

since $\mu(B_0) \leq t\lambda(B_0), \forall t > 0$, where $B_0 := \{x \in B : D(\mu, \lambda, x) = 0\}$. Then the reverse inequality holds.

(iii) "⇒": Apply (i).

"⇐": If $\underline{D}(\mu, \lambda, x) < \infty$ for μ-a.e. x, then, $\forall A$ with $\lambda(A) = 0$, we have

$$\begin{aligned}
\mu(A) &= \lim_{t\to\infty} \mu(\{x \in A : \underline{D}(\mu, \lambda, x) \leq t\}) \\
&\leq \lim_{t\to\infty} t\lambda(A) \\
&= 0.
\end{aligned}$$

□

推论 1.20.1. *Let λ be a Radon measure on $\mathbb{R}^n$. Then*

(i) For any λ-measurable set A, the limit

$$\lim_{r\to 0} \frac{\lambda(A \cap B(x, r))}{\lambda(B(x, r))}$$

equals 1 for λ-a.e. $x \in A$ and 0 for λ-a.e. $x \in A^c$.

(ii) If f is locally λ-integrable then

$$\lim_{r\to 0} \frac{1}{\lambda(B(x, r))} \int_{B(x,r)} f(y)\, \mathrm{d}\lambda(y) = f(x)$$

for λ-a.e. x.

证明. (i) follows from (ii) by setting $f = \chi_A$. For (ii), take $\mu(A) := \int_A f\, \mathrm{d}\lambda$, then $\mu \ll \lambda$. By Theorem 1.20.1(ii),

$$\int_A D(\mu, \lambda, x)\, \mathrm{d}\lambda x = \mu(A) = \int_A f(x)\, \mathrm{d}\lambda x, \quad \forall \lambda\text{-measurable } A.$$

Therefore $f = D(\mu, \lambda, x)$ for λ-a.e. x. □

定义 1.20.3. *Radon measures μ and λ are mutually singular, written $\mu \perp \lambda$, if there exists a Borel set A with $\mu(A) = 0$ and $\lambda(A^c) = 0$.*

例 1.20.1. *(i) If* $\operatorname{supp}\mu \cap \operatorname{supp}\lambda = \varnothing$ *then* $\mu \perp \lambda$.

(ii) Dirac mass δ_0 and Lebesgue measure $\mathcal{L}^n$ are mutually singular. We can take $A = \mathbb{R} \setminus \{0\}$.

定理 1.20.2 (Lebesgue decomposition)**.** *Let μ and λ be finite Radon measures on $\mathbb{R}^n$. There exist unique Borel measures μ_{ac} and μ_s such that*

$$\mu = \mu_{ac} + \mu_s,$$

with $\mu_{ac} \ll \lambda$ and $\mu_s \perp \lambda$. In particular, there exists a Borel function f with

$$\mu(B) = \int_B f \,\mathrm{d}\lambda + \mu_s(B) \qquad \forall \text{ Borel } B.$$

Moreover, $\mu \ll \lambda$ if and only if $\mu_s = 0$.

定理 1.20.3. *Let E be a Borel set with Hausdorff dimension $s \in (1, 2]$. Then for a.e. $e \in S^1$,*

$$\mathcal{H}^1(\pi_e(E)) > 0.$$

证明. Take $\mu \in \mathcal{M}(E)$ with finite 1-energy $I_1(\mu) < \infty$ (see Chapter 2).

$$\begin{aligned}
&\int_{S^1} \underline{D}((\pi_e)_\sharp\mu, \mathcal{L}^1, x) \,\mathrm{d}(\pi_e)_\sharp\mu(x) \,\mathrm{d}e \\
=&\int_{S^1} \int \liminf_{r\to 0} \frac{(\pi_e)_\sharp\mu(B(x,r))}{2r} \,\mathrm{d}(\pi_e)_\sharp\mu(x) \,\mathrm{d}e \\
\le&\liminf_{r\to 0} \frac{1}{2r} \iint |u-v|^{-1} \,\mathrm{d}\mu(u)\mathrm{d}\mu(v) \\
=&cI_1(\mu) \\
<&\infty.
\end{aligned}$$

By Theorem 1.20.1 this implies

$$(\pi_e)_\sharp\mu \ll \mathcal{L}^1 \quad \text{for a.e. } e \in S^1$$

Therefore $0 \neq (\pi_e)_\sharp\mu = f(x)\mathrm{d}x$, and $\mathcal{H}^1(\operatorname{supp}(\pi_e)_\sharp\mu) > 0$. □

1.21　讲义: 几何测度论/ Chapter 4 Fourier Transforms

原网页

The main references of this chapter are [Folland; Real Analysis, Chapter 8], [Mattila; Fourier analysis and Hausdorff dimension, Chapter 3, Chapter 4, Chapter 5], and [Mattila 1987; Spherical averages of Fourier transforms of measures with finite energy; dimensions of intersections and distance sets]

For $f \in L^1$,

$$\hat{f}(\xi) := \int_{\mathbb{R}^n} f(x) e^{-2\pi i x\cdot\xi} \,\mathrm{d}x.$$

For $f \in L^p$: existence of $\hat{f}$?

1.22 Schwarz functions and distributions

定义 1.22.1. *$f \in C^\infty(\mathbb{R}^n)$ is a Schwarz function if $\forall \alpha, \beta \in \mathbb{N}^n$, the semi-norm*

$$\rho_{\alpha,\beta}(f) := \sup_{x\in\mathbb{R}^n} |x^\alpha \partial^\beta f| < \infty.$$

We use $\mathscr{S}(\mathbb{R}^n)$ to denote the class of Schwarz functions.

例 1.22.1. *(i)* $C_0^\infty(\mathbb{R}^n) \subset \mathscr{S}(\mathbb{R}^n)$;

(ii) $e^{-x^2} \in \mathscr{S}(\mathbb{R})$;

(iii) $e^{-|x|} \notin \mathscr{S}(\mathbb{R})$.

(iv) For a polynomial P, and $f \in \mathscr{S}(\mathbb{R}^n)$, we have $Pf \in \mathscr{S}(\mathbb{R}^n)$.

注 1.22.1. *(i) $f \in \mathscr{S}(\mathbb{R}^n)$ if and only if $\forall \alpha \in \mathbb{N}^n, N \in \mathbb{N}$, $\exists C_{\alpha,N}$ such that*

$$|\partial^\alpha f(x)| \le C_{\alpha,N}(1+|x|)^{-N}.$$

(ii) $f \in \mathscr{S} \Rightarrow \partial^\alpha f \in \mathscr{S}$.

(iii) $f_k \to f$ *in* $\mathscr{S}(\mathbb{R}^n) \Leftrightarrow \rho(f_k - f) \to 0, \forall \rho = \rho_{\alpha,\beta}$.

定理 1.22.1. *Let $f, g \in \mathscr{S}(\mathbb{R}^n)$, $y \in \mathbb{R}^n$, $b \in \mathbb{C}$, $\alpha \in \mathbb{N}^n$, $t > 0$. Then*

(i) $\widehat{f+g} = \hat{f} + \hat{g}$,

(ii) $\widehat{bf} = b\hat{f}$,

(iii) $\hat{\tilde{f}}(\xi) = \tilde{\hat{f}}(\xi)$, *where* $\tilde{f}(x) = f(-x)$,

(iv) $\bar{\hat{f}}(\xi) = \hat{\bar{f}}(\xi)$,

(v) $\widehat{\tau^y f}(\xi) = \hat{f}(\xi)e^{-2\pi i y\cdot\xi}$, *where* $\tau^y f(x) = f(x-y)$,

(vi) $[e^{2\pi i y\cdot x}\widehat{f(x)}](\xi) = \hat{f}(\xi - y)$,

(vii) $\hat{f}(x)(0) = \int f$,

(viii) $\left\|\hat{f}\right\|_{L^\infty} \le \|f\|_{L^1}$,

(ix) $\widehat{f * g}(\xi) = \hat{f}(\xi)\hat{g}(\xi)$,

(x) $\widehat{\partial^\alpha f}(\xi) = (2\pi i\xi)^\alpha \hat{f}(\xi)$,

(xi) $\partial^\alpha \hat{f}(\xi) = [\widehat{(-2\pi i x)^\alpha f(x)}](\xi)$,

(xii) $\hat{f} \in \mathscr{S}(\mathbb{R}^n)$,

(xiii) $\widehat{\delta^t f}(\xi) = t^{-n}\hat{f}(t^{-1}\xi) := (\hat{f})_t(\xi)$, *where* $\delta^t f(x) := f(tx)$.

(xiv) $\widehat{f \circ A}(\xi) = \hat{f}(A\xi)$, *where A is an orthogonal matrix.*

定义 1.22.2. *f is radial if $\exists F$ defined on $\mathbb{R}$ such that $f(x) = F(|x|)$.*

推论 1.22.1. *f is radial $\Rightarrow$ $\hat{f}$ is radial.*

Question: under what condition

$$f(x) = \int \hat{f}(\xi)e^{2\pi i x\cdot\xi}\,\mathrm{d}\xi \quad ?$$

定义 1.22.3. *$f^\vee(\xi) := \int f(x)e^{2\pi i x\cdot\xi}\,\mathrm{d}x$ is called inverse Fourier transform.*

定理 1.22.2. *Let $f, g, h \in \mathscr{S}(\mathbb{R}^n)$. Then*

(i) $\int_{\mathbb{R}^n} \hat{f}(x)g(x)\mathrm{d}x = \int_{\mathbb{R}^n} f(x)\hat{g}(x)\mathrm{d}x$
(ii) $(\hat{f})^\vee = f = \widehat{(f^\vee)}$
(iii) $\int f(x)\bar{h}(x)\mathrm{d}x = \int \hat{f}(\xi)\bar{\hat{h}}(\xi)\mathrm{d}\xi$
(iv) $\|f\|_2 = \left\|\hat{f}\right\|_2 = \|f^\vee\|_2$
(v) $\int_{\mathbb{R}^n} f(x)h(x)\mathrm{d}x = \int \hat{f}(x)h^\vee(x)\mathrm{d}x.$

引理 1.22.1. *C_0^∞ is dense in $L^2(\mathbb{R}^n)$.*

$\mathscr{F}$: the extension of the Fourier transformation from $\mathscr{S}$ to L^2. It is an isometry. $\mathscr{F}^{-1}$: extension of $f \mapsto f^\vee$. Clearly $\mathscr{F} \circ \mathscr{F}^{-1} = I$ in $L^2(\mathbb{R}^n)$.

By a standard interpolation one can write $f \in L^p(\mathbb{R}^n), 1 < p < 2$ as $f = f_1 + f_2$ such that $f_i \in L^i(\mathbb{R}^n)$. In this way we define

$$\hat{f} = \hat{f}_1 + \mathscr{F} f_2.$$

命题 1.22.1 (Hausdorff-Young)**.** *Suppose $1 \le p \le 2$, $f \in L^p(\mathbb{R}^n)$. Then*

$$\left\|\hat{f}\right\|_{L^{p'}(\mathbb{R}^n)} \le \|f\|_{L^p(\mathbb{R}^n)},$$

where $\frac{1}{p'} + \frac{1}{p} = 1$.

Linear functional $u : \mathscr{S}(\mathbb{R}^n) \to \mathbb{C}$ is continuous if and only if $\langle \mu, f_k \rangle \to 0$ whenever $f_k \to 0$ in $\mathscr{S}(\mathbb{R}^n)$.

Denote by $\mathscr{S}'(\mathbb{R}^n)$ the set of all continuous linear functionals on $\mathscr{S}(\mathbb{R}^n)$. An element in $\mathscr{S}(\mathbb{R}^n)$ is called a tempered distribution.

命题 1.22.2. *$u \in \mathscr{S}'(\mathbb{R}^n)$ if and only if $\exists C, k, m$ such that*

$$|\langle u, f\rangle| \le C \sum_{\substack{|\alpha|\le m,\\ |\beta|\le k}} \rho_{\alpha,\beta}(f).$$

例 1.22.2. *(i)* $\langle \delta_0, f\rangle = f(0) \le \rho_{0,0}(f), \forall f \in \mathscr{S}(\mathbb{R}^n)$
(ii) $L^p(\mathbb{R}^n) \subset \mathscr{S}'(\mathbb{R}^n), \forall 1 \le p \le \infty$
(iii) $\mathcal{M}(\mathbb{R}^n) \subset \mathscr{S}'(\mathbb{R}^n)$.

定义 1.22.4. *Let $u \in \mathscr{S}'(\mathbb{R}^n)$*
(i) $\partial^\alpha u$ is defined by

$$\langle \partial^\alpha u, f\rangle = (-1)^\alpha \langle u, \partial^\alpha f\rangle,$$

(ii) $\langle \hat{u}, f\rangle := \langle u, \hat{f}\rangle$, $\langle u^\vee, f\rangle := \langle u, f^\vee\rangle$

注 1.22.2. $\hat{\delta}_0 = 1$; $\widehat{(\partial^\alpha \delta_0)}(x) = (2\pi i x)^\alpha$.

定义 1.22.5. *(i)* $\langle \tau^t u, f\rangle := \langle u, \tau^{-t} f\rangle$ $\langle \delta^a u, f\rangle := \langle u, a^{-n}\delta^{a^{-1}} f\rangle$ $\langle \tilde{u}, f\rangle := \langle u, \tilde{f}\rangle$
(ii) For $h \in \mathscr{S}(\mathbb{R}^n)$, $u \in \mathscr{S}'(\mathbb{R}^n)$, define $\langle hu, f\rangle := \langle u, hf\rangle$ $\langle h * u, f\rangle := \langle u, \tilde{h} * f\rangle$.

1.23 Fourier transformation of measures

$\alpha(x,y) := x+y$ Let $u, v \in \mathcal{M}(\mathbb{R}^n)$, then

$$u * v = (\alpha)_\sharp (u \times v).$$

命题 1.23.1. *Let $\mu, \nu \in \mathcal{M}(\mathbb{R}^n)$.*

(i) Convolution of measure is commutative and associative.

(ii) $\forall$ bounded Borel measure function h,

$$\int h \, \mathrm{d}(\mu * \nu) = \iint h(x+y) \, \mathrm{d}\mu(x) \mathrm{d}\mu(y).$$

(iii) $\|\mu * \nu\| = \|\mu\| * \|\nu\|$.

(iv) If $\mathrm{d}\mu = f \, \mathrm{d}\mathcal{L}^n$, and $\mathrm{d}\nu = g \, \mathrm{d}\mathcal{L}^n$, then $(\mu * \nu) = (f * g) \, \mathrm{d}\mathcal{L}^\backslash$.

命题 1.23.2. *Let $1 \le p \le \infty$, $\mu \in \mathcal{M}(\mathbb{R}^n)$, $\forall f \in L^p(\mathbb{R}^n)$, the convolution $f * \mu$ exists for $\mathcal{L}^n$-a.e. x, and*

$$\|f * \mu\|_{L^p(\mathbb{R}^n)} \le \|f\|_{L^p(\mathbb{R}^n)} \|\mu\| . \tag{1.32}$$

证明. It suffices to consider the case $f \ge 0$, when $f * \mu(x)$ is well-defined although it could be ∞. By Minkowski's inequality for integrals, we obtain (1.32), therefore $(f * \mu)(x) < \infty$ for $\mathcal{L}^n$-a.e. x. □

Given $\mu \in \mathcal{M}(\mathbb{R}^n)$, we define

$$\hat{\mu}(\xi) := \int e^{-2\pi i x \cdot \xi} \, \mathrm{d}\mu(x).$$

命题 1.23.3. *If $\mu \in \mathcal{M}(\mathbb{R}^n)$, then*

$$\|\hat{\mu}\|_{L^\infty(\mathbb{R}^n)} \le \|\mu\| ,$$

and

$$|\hat{\mu}(\xi) - \hat{\mu}(\eta)| \le C \|\mu\| \, |\xi - \eta|,$$

where C can be taken as R if $\operatorname{supp}\mu \subset B(0,R)$.

注 1.23.1. *Unlike the case of integrable functions, we do not have $\hat{\mu}(\xi) \to 0$ as $|\xi| \to 0$. For example, $\hat{\delta_0}(\xi) = 1 \notin L^p(\mathbb{R}^n), \forall p < \infty$.*

命题 1.23.4. *Let $f \in L^1(\mathbb{R}^n)$ and $\mu \in \mathcal{M}(\mathbb{R}^n)$. Then*

$$\widehat{f * \mu} = \hat{f}\hat{\mu}.$$

引理 1.23.1. *Let $\mu, \nu \in \mathcal{M}(\mathbb{R}^n)$. If $\hat{\mu} = \hat{\nu}$, then $\mu = \nu$.*

证明. Let $\{\varphi_\varepsilon\}$ be an approximate identity. $\mu_\varepsilon := \varphi_\varepsilon * \mu \to \mu$ weakly. If $\hat{\mu} = 0$, then $\hat{\mu}_\varepsilon = \hat{\varphi}_\varepsilon \hat{\mu} = 0$. Recalling that $\mu_\varepsilon \in C_c^\infty$, we have $\mu_\varepsilon = 0$. As a result, $\mu = 0$. □

Given $\mu \in \mathcal{M}(\mathbb{R}^n)$ we $\mu \ll \mathcal{L}^n$, we can find $f \in L^1(\mathbb{R}^n)$ such that $\mathrm{d}\mu = f \, \mathrm{d}\mathcal{L}^n$. If there is no risk of confusion, we identify μ and f.

定理 1.23.1. *Let $\mu \in \mathcal{M}(\mathbb{R}^n)$. If $\hat{\mu} \in L^2(\mathbb{R}^n)$, then $\mu \in L^2(\mathbb{R}^n)$.*

证明. As the Fourier transform is an isometry on $L^2(\mathbb{R}^n)$, we can find $f \in L^2(\mathbb{R}^n)$ such that $\hat{f} = \hat{\mu}$. This implies that $\hat{\mu}_\varepsilon = \hat{f}_\varepsilon$ where $f_\varepsilon = f * \varphi_\varepsilon$. Since $f_\varepsilon \to f$ in $L^2(\mathbb{R}^n)$, we have $f_\varepsilon \to f$ weakly. Recalling that $\mu_\varepsilon \to \mu$ weakly, we have $\mu = f$. □

1.24 Energy and Fourier transformation

$0 < s < n$ s-energy $I_s(\mu) := \iint |x-y|^{-s}\,\mathrm{d}\mu(x)\mathrm{d}\mu(y)$. $k_s(x) := |x|^{-s}$ Riesz kernel.

定理 1.24.1. *Let $\mu \in \mathcal{M}(\mathbb{R}^n)$, and $0 < s < n$. Then*

$$I_s(\mu) = \gamma(n,s)\int |\hat{\mu}(\xi)|^2 |\xi|^{s-n}\,\mathrm{d}\xi. \tag{1.33}$$

定理 1.24.2. *Let $0 < s < n$. There exists $\gamma(n,s) \in (0,\infty)$ such that*

$$\hat{k}_s = \gamma(n,s)k_{n-s}$$

in the sense of tempered distributions.

证明. As $(\widehat{e^{-\pi|x|^2}})(\xi) = e^{-\pi|\xi|^2}$, we have, for $t > 0$,

$$\int_{\mathbb{R}^n} e^{-\pi t|x|^2}\varphi(x)\,\mathrm{d}x = t^{-n/2}\int_{\mathbb{R}^n} e^{-\pi|x|^2/t}\hat{\varphi}(x)\,\mathrm{d}x. \tag{1.34}$$

We multiply both sides of (1.34) by $t^{-s/2-1}$ and integrate over $t \in (0,\infty)$, then

$$\int_{\mathbb{R}^n}\int_0^\infty e^{-\pi t|x|^2}\cdot t^{-s/2-1}\,\mathrm{d}t\;\varphi(x)\,\mathrm{d}x = \int_{\mathbb{R}^n}\int_0^\infty e^{-\pi|x|^2/t}\cdot t^{-n/2+s/2-1}\,\mathrm{d}t\;\hat{\varphi}(x)\,\mathrm{d}x. \tag{1.35}$$

Recall that $\Gamma(\lambda) := \displaystyle\int_0^\infty e^{-t}t^{\lambda-1}\,\mathrm{d}t$, have

$$\int_0^\infty e^{-\pi t|x|^2}t^{s/2-1}\,\mathrm{d}t = \Gamma\left(\frac{s}{2}\right)(\pi|x|^2)^{-s/2},$$

and

$$\int_0^\infty e^{-\pi t^{-1}|x|^2}t^{-\frac{n-s}{2}}\frac{\mathrm{d}t}{t} = \Gamma\left(\frac{n-s}{2}\right)(\pi|x|^2)^{\frac{s-n}{2}}.$$

Plugging them into (1.35), we obtain

$$\int_{\mathbb{R}^n}|x|^{-s}\varphi(x)\,\mathrm{d}x = \frac{\Gamma\left(\frac{n-s}{2}\right)\pi^{\frac{2s-n}{2}}}{\Gamma\left(\frac{s}{2}\right)}\int_{\mathbb{R}^n}|x|^{-\frac{n-s}{2}}\hat{\varphi}(x)\,\mathrm{d}x.$$

$$(\mu*\nu)(E) := \iint \chi_E(x+y)\,\mathrm{d}\mu(x)\mathrm{d}\nu(y)$$

$$\begin{aligned}\iint k_s(x-y)\,\mathrm{d}\mu(x)\mathrm{d}\mu(y) &= \int (k_s*\mu)(y)\,\mathrm{d}\mu(y)\\ &= (\widehat{k_s*\mu})(\xi)\bar{\hat{\mu}}(\xi)\,\mathrm{d}\xi\\ &= \hat{k_s}(\xi)\hat{\mu}(\xi)\bar{\hat{\mu}}(\xi)\,\mathrm{d}\xi\\ &= \gamma(n,s)\int|\xi|^{s-n}|\hat{\mu}(\xi)|^2\,\mathrm{d}\xi\end{aligned}$$

$$\iint |x-y|^{-s}\,\mathrm{d}\mu(x)\mathrm{d}\mu(y) = \gamma(n,s)\int|\xi|^{s-n}|\hat{\mu}(\xi)|^2\,\mathrm{d}\xi. \tag{1.36}$$

□

Proof of Theorem 1.24.1. Equation (1.36) is easy when $\mathrm{d}\mu = \varphi\,\mathrm{d}x$ with $\varphi \in \mathscr{S}(\mathbb{R}^n)$. Hence for $\mu_\varepsilon := \varphi_\varepsilon * \mu$, we have

$$\begin{aligned}&\iint \left(\iint |x-y|^{-s}\varphi_\varepsilon(x-z)\varphi_\varepsilon(y-w)\,\mathrm{d}x\mathrm{d}y\right)\,\mathrm{d}\mu(z)\mathrm{d}\mu(w)\\ =&I_s(\mu_\varepsilon)\\ =&\gamma(n,s)\int |\hat{\mu}(\xi)\hat{\varphi}(\varepsilon\xi)|^2|\xi|^{s-n}\,\mathrm{d}\xi.\end{aligned}$$

Claim 1:

$$\lim_{\varepsilon\to 0}\int |\hat{\mu}(\xi)\hat{\varphi}(\varepsilon\xi)|^2|\xi|^{s-n}\,\mathrm{d}\xi = \int |\hat{\mu}(\xi)|^2|\xi|^{s-n}\,\mathrm{d}\xi. \tag{1.37}$$

Claim 2: For $z \neq w$, we have

$$\lim_{\varepsilon\to 0}\iint |x-y|^{-s}\varphi_\varepsilon(x-z)\varphi_\varepsilon(y-w)\,\mathrm{d}x\mathrm{d}y = |z-w|^{-s}, \tag{1.38}$$

and

$$\left|\iint |x-y|^{-s}\varphi_\varepsilon(x-z)\varphi_\varepsilon(y-w)\,\mathrm{d}x\mathrm{d}y\right| \lesssim |z-w|^{-s} \tag{1.39}$$

uniformly in $\varepsilon > 0$.

By **Claim 1** and **Claim 2**, we complete the proof by Lebesgue's Dominated Convergence Theorem, and Fatou's Lemma.

Proof of Claim 1: RHS of (1.37) $< \infty$: LDCT. RHS of (1.37) $= \infty$: Fatou's Lemma.

Proof of Claim 2: LHS of (1.38)

$$= \int |\varepsilon(u-v)+(z-w)|^{-s}\varphi(u)\varphi(v)\,\mathrm{d}u\mathrm{d}v. \tag{1.40}$$

As $z \neq w$, we have $|\varepsilon(u-v)+(z-w)| \sim |z-w|$ for ε small. We can apply LDCT to verify (1.38). Fix $\varepsilon > 0$. Let $a = z-w$, by (1.40), (1.39) is obvious when $\varepsilon \geq 10|a|$ or $\varepsilon \leq \frac{|a|}{10}$. It remains to consider the case $\frac{|a|}{10} \leq \varepsilon|u-v| \leq 10|a|$, which implies that $|a| \leq 20\varepsilon$. Using this and decomposing (u,v) dyadically (in terms of $\left|u-v+\frac{a}{\epsilon}\right|$), we obtain (1.39). □

If $\mu \in \mathcal{M}(\mathbb{R}^n)$ and $I_s(\mu) < \infty$, where $0 < s < n$, then

$$\begin{aligned}\int_{B(0,R)} |\hat{\mu}(x)|^2\,\mathrm{d}x &\leq R^{n-s}\int_{B(0,R)} |x|^{s-n}|\hat{\mu}(x)|^2\,\mathrm{d}x\\ &\leq CI_s(\mu)R^{n-s}.\end{aligned} \tag{1.41}$$

引理 1.24.1. *Let $0 < s < n$, and let μ be an s-Frostman measure on $\mathbb{R}^n$. Then $\exists C \in (0,\infty)$ such that*

$$\int_{B(0,R)} |\hat{\mu}(x)|^2\,\mathrm{d}x \leq CR^{n-s}.$$

证明. Take $\varphi \in \mathscr{S}(\mathbb{R}^n)$ such that $\hat{\varphi} \geq 0$ and $\hat{\varphi}(x) = 1$ for $|x| \leq 1$. Define $\varphi_R(x) = R^n\varphi(Rx)$. Then

$$\begin{aligned}\int_{B(0,R)} |\hat{\mu}(x)|^2\,\mathrm{d}x &\leq \int \hat{\varphi}_R|\hat{\mu}(x)|^2\,\mathrm{d}x\\ &= \int \hat{\varphi}_R(x)\hat{\mu}(x)\bar{\hat{\mu}}(x)\,\mathrm{d}x\\ &= \int (\varphi_R * \mu)(x)\,\mathrm{d}\mu(x)\end{aligned}$$

By dyadic decomposition,

$$\varphi_R * \mu \lesssim R^n \mu(B(0, R^{-1})) \sim R^{n-s}.$$

Therefore,

$$\int_{B(0,R)} |\hat{\mu}(x)|^2 \, \mathrm{d}x \lesssim R^{n-s}.$$

□

The following lemma says that the "converse" of (1.41) is true.

引理 1.24.2. *Let $0 < s < n$. Suppose $\mu \in \mathcal{M}(\mathbb{R}^n)$ satisfies*

$$\int_{B(0,R)} |\hat{\mu}(x)|^2 \, \mathrm{d}x \leq CR^{n-s} \quad \forall R \geq 1, \tag{1.42}$$

then $I_t(\mu) < \infty$, $\forall t \in (0, s)$. In particular, $\dim(\operatorname{supp} \mu) \geq s$.

证明. Recall that

$$I_t(\mu) = c \int_{\mathbb{R}^n} |\xi|^{t-n} |\hat{\mu}(\xi)| \, \mathrm{d}\xi,$$

we will control the RHS:

$$\int_{B(0,1)} |\xi|^{t-n} |\hat{\mu}(\xi)|^2 \, \mathrm{d}\xi \leq C,$$

$$\begin{aligned}
\int_{|\xi| \geq 1} |\xi|^{t-n} |\hat{\mu}(\xi)|^2 \, \mathrm{d}\xi &= \sum_{k \geq 1} \int_{|\xi| \sim 2^k} |xi|^{t-n} |\hat{\mu}(\xi)|^2 \, \mathrm{d}\xi \\
&\leq \sum_{k \geq 1} 2^{k(t-n)} \int_{B(o, 2^k)} |\hat{\mu}(\xi)|^2 \, \mathrm{d}\xi \\
&\leq \sum_{k \geq 1} 2^{k(t-n)} 2^{k(n-s)} \\
&< \infty.
\end{aligned}$$

□

注 1.24.1. $\mu(B(x,r)) \leq Cr^s \implies \displaystyle\int_{B(0,R)} |\hat{\mu}(\xi)|^2 \, \mathrm{d}\xi \leq CR^{n-s} \implies \dim_{\mathcal{H}}(\operatorname{supp} \mu) \geq s.$ (1.42) *morally says that*

$$|\hat{\mu}(\xi)| \leq C|\xi|^{-\frac{s}{2}}. \tag{1.43}$$

例 1.24.1. *$E := [0,1] \times \{0\} \subset \mathbb{R}^2$ has Hausdorff dimension 1, which does not satisfy (1.38) for any $s > 0$. Let μ be the 1-dimensional Hausdorff measure on E, then $\hat{\mu}(\xi_1, \xi_2)$ is a function that depends only on ξ_1, which cannot go to 0 as $|\xi| \to \infty$.*

定义 1.24.1. *The **Fourier Dimension** of a set $A \subset \mathbb{R}^n$ is defined by*

$$\dim_F(A) := \sup \left\{ s \geq 0 : \exists \mu \in \mathcal{M}(A) \text{ s.t. } |\hat{\mu}(\xi)| \leq C|\xi|^{-\frac{s}{2}}, \forall \xi \in \mathbb{R}^n \right\}.$$

定理 1.24.3. *Suppose A is a Borel set. Then*

$$\dim_F(A) \leq \dim_{\mathcal{H}}(A).$$

证明. Apply Lemma 1.24.2. □

定义 1.24.2. *A set $A \subset \mathbb{R}^n$ is called salem set if* $\dim_F(A) = \dim_{\mathcal{H}}(A)$.

注 1.24.2. *It is equivalent to say,* $\forall s < \dim_{\mathcal{H}} A$, $\exists \mu \in \mathcal{M}(A)$ *such that*

$$|\hat{\mu}(\xi)| \lesssim |\xi|^{-\frac{s}{2}}.$$

例 1.24.2. *(i) Let $n \geq 2$, and let σ be the surface measure on the unit sphere S^{n-1}. It is well known that*

$$|\hat{\sigma}(\xi)| \leq C(1+|\xi|)^{-\frac{n-1}{2}}.$$

Thus S^{n-1} is a salem set.

(ii) Fix $\alpha > 0$, and define

$$E_\alpha := \left\{ x \in \mathbb{R} : \exists \text{ infinitely many rationals } \frac{p}{q} \text{ with } q \geq 1 \text{ s.t. } \left| x - \frac{p}{q} \right| \leq q^{-(2+\alpha)} \right\}.$$

Then E_α is a salem set with Hausdorff dimension $\frac{2}{2+\alpha}$.

1.25 Orthogonal Projections

定理 1.25.1. *Let E be a Borel set in $\mathbb{R}^2$ with dimension s. If $s \leq 1$, then*

$$\dim(\pi_e(E)) = s \qquad \text{for a.e. } e \in S^1;$$

if $s > 1$, then

$$\mathcal{H}^1(\pi_e(E)) > 0 \qquad \text{for a.e. } e \in S^1.$$

证明. Let $\mu_e := (\pi_e)_\sharp \mu$. $s \leq 1$: Take $\mu \in \mathcal{M}(E)$ satisfying $I_t(\mu) < \infty$ with $t < s$. $\hat{\mu}_e(r) \int e^{irt} \, \mathrm{d}\mu_e(t) = \hat{\mu}(re)$. Then

$$\begin{aligned}
&\int_{S^1} I_t(\mu_e) \, \mathrm{d}\sigma(e) \\
=&\int_{S^1} \iint |u-v|^{-t} \, \mathrm{d}\mu_e(u) \mathrm{d}\mu_e(v) \mathrm{d}\sigma(e) \\
=&\int_{S^1} \iint |e \cdot (x-y)|^{-t} \, \mathrm{d}\mu(x) \mathrm{d}\mu(y) \mathrm{d}\sigma(e) \\
=&\iint |x-y|^{-t} \int_{S^1} \left| e \cdot \frac{x-y}{|x-y|} \right|^{-t} \mathrm{d}\sigma(e) \mathrm{d}\mu(x) \mathrm{d}\mu(y) \\
\leq& C_t I_t(\mu).
\end{aligned}$$

(因为 $\int_{S^1} |e \cdot e_1|^{-t} \, \mathrm{d}\sigma(e) \leq C_t < \infty$ 当 $t < 1$ 时) $I_t(\mu_e) < \infty$ for a.e. e, $\dim(\pi_e(E)) \geq t$ for a.e. e, $\dim(\pi_e(E)) = s$ for a.e. e.

$s > 1$: Take μ such that $I_1(\mu) < \infty$.

$$\begin{aligned}
&\int_{S^1} I_1(\mu_e) \, \mathrm{d}\sigma(e) \\
=&\int_{S^1} \int |\hat{\mu}(re)|^2 \, \mathrm{d}r \mathrm{d}\sigma(e) \\
=&\int |\xi| |\hat{\mu}(\xi)|^2 \, \mathrm{d}\xi = I_1(\mu) < \infty.
\end{aligned}$$

□

定理 1.25.2. *Let $A \subset \mathbb{R}^2$ be a Borel set with $\dim A = s \leq 1$. Then, $\forall t \in [0, s]$, the exceptional set*

$$E_t := \{e \in S^1 : \dim \pi_e(E) < t\}$$

satisfies

$$\dim E_t \leq t.$$

证明. Fixing $\sigma < t$, take $\mu \in \mathcal{M}(A)$ with $I_\sigma(\mu) < \infty$. As $\mu_e \in \mathcal{M}(\pi_e(A))$, we have

$$E_\sigma \subset E_\sigma^1 := \{e \in S^1 : I_\sigma(\mu_e) = \infty\}.$$

Actually, $\forall e \in E_\sigma$, $I_\sigma(\mu_e) = \infty$. Otherwise, $\dim(\operatorname{supp} \mu_e) \geq \sigma$, which contradicts that $\dim \pi_e(A) < \sigma$. Suppose $\dim E_\sigma^1 > t$, then we can find a (non-zero) t-Frostman measure $\nu \in \mathcal{M}(E_\sigma^1)$. Observe that

$$\begin{aligned}
&\int_{S^1} I_\sigma(\mu_e) \, \mathrm{d}\nu(e) \\
=&\iint \int_{S^1} |e \cdot (x-y)|^{-\sigma} \, \mathrm{d}\nu(e) \mathrm{d}\mu(x) \mathrm{d}\mu(y) \\
\leq& C I_\sigma(\mu) < \infty
\end{aligned}$$

as

$$\begin{aligned}
\int_{S^1} |e \cdot e_1|^{-\sigma} \, \mathrm{d}\nu(e) &\lesssim \sum_{k \geq 1} (2^{-k})^{-\sigma} \nu(I_k) \\
&\leq C_\nu \sum_{k \geq 1} 2^{k\sigma} 2^{-kt} \\
&\leq C_\nu.
\end{aligned}$$

This contradicts with

$$\int_{S^1} I_\sigma(\mu_e) \, \mathrm{d}\nu(e) = \infty.$$

Therefore $\dim E_\sigma^1 \leq t$. □

1.26 Falconer's distance set conjecture

Given a finite subset $P \subset \mathbb{R}^2$ of N points, we define the associated distance set

$$\Delta(P) := \{|x - y| : x, y \in P\}.$$

命题 1.26.1. *Let $P = \{(m, n) \in \mathbb{Z}^2 : 0 \leq m, n \leq \sqrt{N}\}$, then* $\left|\Delta(P) \geq C \dfrac{N}{\sqrt{\log N}}\right|$.

conjecture (Distinct Distance)**.** *Suppose that $P \subset \mathbb{R}^2$ is a set of N points. Then*

$$|\Delta(P)| \geq C \frac{N}{\sqrt{\log N}}.$$

Guth-Katz 2015: $|\Delta(P)| \geq C \dfrac{N}{\log N}$.

$$u_2(N) := \sup_{|P|=N} |\{(x, y) \in P \times P : |x - y| = 1\}|.$$

conjecture (Unit Distance)**.** $u_2(N) \le CN\sqrt{\log N}$.

注 **1.26.1.** *Conjecture 2* $\Longrightarrow$ *Conjecture 1. In fact,*

$$\begin{aligned} N^2 &= \#\{(x,y) : x,y \in P\} \\ &= \sum_{d\in\Delta(P)} \sum 1 \\ &\le |\Delta(P)| u_2(N). \end{aligned}$$

Now replace P with arbitrary Borel set $E \subset \mathbb{R}^2$, $\Delta(E) := \{|x-y| : x,y \in E\} \subset \mathbb{R}$. Is $\dim(\Delta(E)) = \min\{\dim E, 1\}$ true?

conjecture (Falconer)**.** *Suppose $E \subset \mathbb{R}^n$ is Borel with* $\dim E > \frac{n}{2}$. *Then*

$$|\Delta(E)| > 0.$$

In particular,

$$\dim \Delta(E) = 1.$$

定理 **1.26.1.** *If* $\dim E > \frac{n+1}{2}$*, then the interior of $\Delta(E)$ is non-empty. In particular, $|\Delta(E)| > 0$.*

To prove **Theorem 1.26.1**, we introduce some notations first. $\pi(x,y) := |x-y|$, $\pi : E\times E \to \Delta(E)$, π introduces a measure on $\mathbb{R}$, the **push-forward measure** of $\mu \times \mu$,

$$\begin{aligned} \delta_\mu &:= (\pi)_\sharp(\mu\times\mu), \\ \delta_\mu(B) &= (\mu\times\mu)(\pi^{-1}B). \end{aligned}$$

It's clear that

$$\langle \delta_\mu, \varphi\rangle = \iint_{E\times E} \varphi(|x-y|)\, \mathrm{d}\mu(x)\mathrm{d}\mu(y), \quad \forall \varphi \in C_c(\mathbb{R}),$$

therefore $\operatorname{supp}\delta_\mu \subset \Delta(E)$. Denote σ_r for the Hausdorff measure on $S_r^{n-1} := \{x \in \mathbb{R}^n : |x| = r\}$. Let $\varphi_r(x) := r^{n-1}\varphi(rx)$, then

$$\langle \sigma_r, \varphi\rangle = \langle \sigma, \varphi_r\rangle,$$

and

$$\hat{\sigma}_r(\xi) = \hat{\sigma}(r\xi) r^{n-1}.$$

引理 **1.26.1.** $\hat{\sigma}(\xi) = \dfrac{2\pi}{|\xi|^{\frac{n-2}{2}}} J_{\frac{n-2}{2}}(2\pi|\xi|)$*, where the Bessel function J_α of order α is defined by*

$$J_\alpha(t) := \frac{\left(\frac{t}{2}\right)^\alpha}{\Gamma\left(\alpha+\frac{1}{2}\right)\Gamma\left(\frac{1}{2}\right)} \int_{-1}^{1} e^{its}(1-s^2)^{\alpha-1/2}\, \mathrm{d}s$$

for $\operatorname{Re}\alpha > -\frac{1}{2}, t \ge 0$*. In particular, $|\hat{\sigma}(\xi)| \le C(1+|\xi|)^{-\frac{n-1}{2}}$.*

引理 **1.26.2.** *If $\mu \in \mathcal{M}(\mathbb{R}^n)$ satisfies $I_{\frac{n+1}{2}}(\mu) < \infty$, then δ_μ is a continuous function defined by*

$$\delta_\mu(r) = r^{n-1} \int_{\mathbb{R}^n} \hat{\sigma}(r\xi)|\hat{\mu}(\xi)|^2\, \mathrm{d}\xi.$$

证明. We consider first the case when $\mathrm{d}\mu = f\mathrm{d}\mathcal{L}^n$, where $f \in C_c(\mathbb{R}^n)$ is non-negative and real-valued. For $g \in C_c(\mathbb{R})$,

$$\begin{aligned}\langle \delta_f, g\rangle &= \iint g(|x-y|)f(x)f(y)\,\mathrm{d}x\mathrm{d}y\\ &= \iint g(|z|)f(y+z)\,\mathrm{d}z\,f(y)\,\mathrm{d}y\\ &= \int_0^\infty \int_{S^{n-1}} g(r)f(g+r\theta)\,\mathrm{d}\sigma(\theta)\,r^{n-1}\,\mathrm{d}r\,f(y)\,\mathrm{d}y\\ &= \int_0^\infty g(r)\int f(y)\int_{S^{n-1}} f(y+r\theta)r^{n-1}\,\mathrm{d}\sigma(\theta)\mathrm{d}y\mathrm{d}r.\end{aligned}$$

Hence

$$\begin{aligned}\delta_f(r) &= \int (f*\sigma_r)(y)f(y)\,\mathrm{d}y\\ &= \int \hat{f}(\xi)\hat{\sigma}_r(\xi)\bar{\hat{f}}(\xi)\,\mathrm{d}\xi\\ &= r^{n-1}\int \hat{\sigma}(r\xi)|\hat{f}(\xi)|^2\,\mathrm{d}\xi.\end{aligned}$$

Take a real-valued $\psi \in C_c^\infty(\mathbb{R}^n)$ with $\int \psi = 1$, and define $\psi_\varepsilon(z) = \varepsilon^{-n}\psi(\varepsilon^{-1}z)$. Let $\mu_\varepsilon = \mu * \psi_\varepsilon$, then $\mu_\varepsilon \to \mu$. In particular, $\hat{\mu}_\varepsilon(\xi) \to \hat{\mu}(\xi), \forall \xi$. Since $|\hat{\sigma}(r\xi)| \le c(1+|r\xi|)^{-\frac{n-1}{2}}$, and

$$I_{\frac{n+1}{2}} = \int |\xi|^{-\frac{n-1}{2}}|\hat{\mu}(\xi)|^2\,\mathrm{d}\xi,$$

we obtain from Lebesgue's dominated convergence theorem that

$$\begin{aligned}\delta_{\mu_\varepsilon}(r) &= r^{n-1}\int \hat{\sigma}(r\xi)|\hat{\mu}_\varepsilon(\xi)|^2\,\mathrm{d}\xi\\ &\to r^{n-1}\int \hat{\sigma}(r\xi)|\mu(\xi)|^2\,\mathrm{d}\xi.\end{aligned}$$

We show below that $\delta_{\mu_\varepsilon} \rightharpoonup \delta_\mu$. As

$$\begin{aligned}&\int F\,\mathrm{d}(\mu_\varepsilon \times \mu_\varepsilon)\\ =&\int \hat{F}(\xi_1,\xi_2)\hat{\mu}(\xi_1)\hat{\psi}(\varepsilon\xi_1)\hat{\mu}(\xi_2)\hat{\psi}(\varepsilon\xi_2)\,\mathrm{d}\xi_1\mathrm{d}\xi_2\\ \to&\int \hat{F}(\xi_1,\xi_2)\hat{\mu}(\xi_1)\hat{\mu}(\xi_2)\,\mathrm{d}\xi_1\mathrm{d}\xi_2\\ =&\int F\,\mathrm{d}(\mu\times\mu),\end{aligned}$$

we have $\mu_\varepsilon \times \mu_\varepsilon \rightharpoonup \mu \times \mu$. This implies that

$$\delta_{\mu_\varepsilon} \rightharpoonup \delta_\mu$$

using that $\delta_\mu = (\pi)_\sharp(\mu\times\mu)$. □

Proof of Theorem 1.26.1. Take $\mu \in \mathcal{M}(E)$ with $I_{\frac{n+1}{2}}(\mu) < \infty$. Applying **Lemma 1.26.2**, we obtain δ_μ which is a non-zero continuous function. In particular, $\operatorname{supp}\mu$ contains an interior point, then $\Delta(F)$ contains nonempty interior. □

1.27 Spherical average

$\sigma(\mu)(r) = \int_{S^{n-1}} |\hat{\mu}(r\theta)|^2 \, \mathrm{d}\sigma(\theta)$ spherical average.

引理 1.27.1. *For all $\mu \in \mathcal{M}(\mathbb{R}^n)$ with $I_{\frac{n+1}{2}}(\mu) < +\infty$,*

$$\delta_\mu(t) = ct^{n/2} \int_0^\infty r^{n/2} J_{\frac{n-2}{2}}(tr)\sigma(\mu)(r) \, \mathrm{d}r.$$

证明.

$$\begin{aligned}
\delta_\mu(t) &= \int \hat{\sigma}_t(\xi)|\hat{\mu}(\xi)|^2 \, \mathrm{d}\xi \\
&= \int_0^\infty t^{n-1}\hat{\sigma}(tr) \int_{S^{n-1}} |\hat{\mu}(r\theta)|^2 \, \mathrm{d}\sigma(\theta) \, r^{n-1} \, \mathrm{d}r \\
&= \int_0^\infty (tr)^{n/2} \cdot J_{\frac{n-2}{2}}(2\pi tr)\sigma(\mu)(r) \, \mathrm{d}r.
\end{aligned}$$

□

定义 1.27.1. $\Delta_\mu(t) := t^{-\frac{n-1}{2}}\delta_\mu(t)$, $\Sigma(\mu)(r) = r^{\frac{n-1}{2}}\sigma(\mu)(r)$.

引理 1.27.2. *Let $\mathcal{H}_k$ be the Hankel transformation*

$$\mathcal{H}_k(f)(t) = \int_0^\infty J_k(tr)f(r)r \, \mathrm{d}r,$$

then

$$\mathcal{H}_k\mathcal{H}_k(f) = f.$$

命题 1.27.1. *Let $f \in C_c(\mathbb{R}^n)$ be non-negative and real-valued. Then*

$$\int_0^\infty \Delta_f(t)^2 \, \mathrm{d}t = c\int_0^\infty \Sigma(f)(r)^2 \, \mathrm{d}r.$$

证明. By **Lemma 1.27.1**,

$$\begin{aligned}
&\int_0^\infty \Delta_f(t)^2 \, \mathrm{d}t \\
=&c\int_0^\infty t \int_0^\infty r^{1/2} J_{\frac{n-2}{2}}(2\pi tr)\Sigma(f)(r) \, \mathrm{d}r \\
&\qquad \int_0^\infty s^{1/2} J_{\frac{n-2}{2}}(2\pi ts)\Sigma(f)(s) \, \mathrm{d}s\mathrm{d}t \\
=&c\int_0^\infty r^{1/2}\Sigma(f)(r) \int_0^\infty tJ_{\frac{n-2}{2}}(2\pi tr) \int_0^\infty sJ_{\frac{n-2}{2}}(2\pi ts)\Sigma(f)(s)s^{-1/2} \, \mathrm{d}s\mathrm{d}t\mathrm{d}r \\
=&c\int_0^\infty r^{1/2}\Sigma(f)(r) \int_0^\infty tJ_{\frac{n-2}{2}}(tr)\mathcal{H}_{\frac{n-2}{2}}(\Sigma(f)(s)s^{-1/2})(t) \, \mathrm{d}t\mathrm{d}r \\
=&c\int_0^\infty r^{1/2}\Sigma(f)(r) \cdot \Sigma(f)(r)r^{-1/2} \, \mathrm{d}r \\
=&c\int_0^\infty \Sigma(f)(r)^2 \, \mathrm{d}r.
\end{aligned}$$

□

球面平均与 Energy 有天然的联系:

引理 1.27.3. $I_\alpha(\mu)=\int_0^\infty r^{\alpha-1}\sigma(\mu)(r)\,\mathrm{d}r.$

证明. $I_\alpha(\mu)=\int_{\mathbb{R}^n}|\xi|^{\alpha-n}|\hat{\mu}(\xi)|^2\,\mathrm{d}\xi=\int_0^\infty r^{\alpha-1}\int_{S^{n-1}}|\hat{\mu}(r\theta)|^2\,\mathrm{d}\sigma(\theta)\mathrm{d}r.$ □

定理 1.27.1. *Let $C_\gamma:=\sup_{r>0}\sigma(\mu)(r)r^\gamma$. For $\mu\in\mathcal{M}(\mathbb{R}^n)$, we have*

$$\int_0^\infty \Delta_\mu(t)^2\,\mathrm{d}t\lesssim I_\alpha(\mu)C_{n-\alpha}.$$

Suppose that, for some $\alpha\in\left(\frac{n}{2},n\right)$, we have

$$I_\alpha(\mu)<\infty\implies C_\gamma<\infty$$

for some $\gamma>n-\alpha$. Then $\Delta(\mu)\in L^2$ for all $\mu\in\mathcal{M}(\mathbb{R}^n)$ satisfying $I_\alpha(\mu)<\infty$. In particular, Falconer's distance set conjecture is true for α.

证明. By **Proposition 1.27.1** and **Lemma 1.27.3**,

$$\begin{aligned}\int_0^\infty|\Delta_\mu(t)|^2\,\mathrm{d}t&=\int_0^\infty r^{n-1}\sigma(\mu)(r)^2\,\mathrm{d}r\\&\le C_{n-\alpha}\int_0^\infty r^{\alpha-1}\sigma(\mu)(r)\,\mathrm{d}r\\&\sim C_{n-\alpha}I_\alpha(\mu).\end{aligned}$$

As $\sigma(\mu)(r)<\infty$, if $C_\gamma<\infty$, then

$$C_\beta<\infty\quad\text{for}\quad\beta\le r.$$

Then "$C_\gamma<\infty$ with $\gamma>n-\alpha$" implies $C_{n-\alpha}<\infty$, which implies $\Delta_\mu\in L^2$. □

1.28　讲义: 几何测度论/ Chapter 5 Self-similar Sets

原网页

The main references of this chapter are [Hochman; Lectures on fractal geometry and dynamics, Chapter 5], [Bishop, Peres; Fractals in Probability and Analysis, Chapter 2], [Mattila; Fourier analysis and Hausdorff dimension, Chapter 9], and [Shen; Lectures on Bernoulli Convolution]

1.29　Hausdorff metric

Let (X,d) be a metric space. $\forall\varepsilon>0$, we define

$$A^{(\varepsilon)}:=\{x\in X:d(x,A)<\varepsilon\}.$$

we say that A is ε-dense in B if $B\subset A^{(\varepsilon)}$.

Definition Let $\mathrm{Cpt}(X)$ be the collection of compact non-empty subsets of X. We define the Hausdorff metric between A and B by

$$d_{\mathcal{H}}(A,B):=\inf\{\varepsilon>0:A\subset B^{(\varepsilon)},B\subset A^{\varepsilon}\}.$$

Example $A=\left\{\frac{1}{2}\right\},B=[0,1]$. Then $d(A,B)=0$, $d_{\mathcal{H}}(A,B)=\frac{1}{2}$.

引理 1.29.1. *Let $\varepsilon < \delta$. Then $\overline{A^{(\varepsilon)}} \subset A^{(\delta)}$.*

证明. Exercise. □

定理 1.29.1. *Let $E \subset X$. Then the followings are equivalent:*

(i) E is complete and totally bounded.

(ii) Every subsequence in E has a subsequence convergent to a point in E.

(iii) E is compact.

(Remark: E is defined to be totally bounded if, for all $\varepsilon > 0$, E can be covered by finitely many balls of radius ε.)

定理 1.29.2. *Let (X, d) be a metric space.*

(i) $d_{\mathcal{H}}$ is a metric on $\mathrm{Cpt}(X)$.

(ii) If $\{A_n\} \subset \mathrm{Cpt}(X)$ and $A_1 \supset A_2 \supset \ldots$, then

$$A_n \xrightarrow{d_{\mathcal{H}}} \bigcap_{n=1}^{\infty} A_n.$$

(iii) If (X, d) is complete, then $(\mathrm{Cpt}(X), d_{\mathcal{H}})$ is complete.

(iv) If $A_n \xrightarrow{d_{\mathcal{H}}} A$, then A is the set of limits

$$a = \lim_{n\to\infty} a_n$$

of convergent sequence $a_n \in A_n$.

(v) (X, d) is totally bounded $\implies$ $(\mathrm{Cpt}(X), d_{\mathcal{H}})$ is totally bounded.

(vi) (X, d) is compact $\implies$ $(\mathrm{Cpt}(X), d_{\mathcal{H}})$ is compact.

证明. (i) By definition,

$$d_{\mathcal{H}} \geq 0, \qquad d_{\mathcal{H}}(A, B) = d_{\mathcal{H}}(B, A).$$

Suppose $A \neq B$. WLOG, we assume $\exists x \in A \setminus B$. Since B is compact, we have

$$d(x, B) = \delta > 0.$$

This implies

$$x \notin B^{(\varepsilon)}, \qquad \forall \varepsilon < \delta.$$

As a result, $d_{\mathcal{H}}(A, B) \geq \delta > 0$. For the triangle inequality, suppose $d_{\mathcal{H}}(A, B) = d_1$, $d_{\mathcal{H}}(B, C) = d_2$. Then, $\forall \varepsilon > 0$,

$$A \subset B^{(d_1+\varepsilon)}, \quad B \subset C^{(d_2+\varepsilon)}.$$

This implies that $A \subset C^{(d_1+d_2+2\varepsilon)}$. Similarly, $C \subset A^{(d_1+d_2+2\varepsilon)}$. Therefore

$$d_{\mathcal{H}}(A, C) \leq d_1 + d_2 = d_{\mathcal{H}}(A, B) + d_{\mathcal{H}}(B, C).$$

(ii) Let $A := \bigcap_{n=1}^{\infty} A_n \subset A_n$. It suffices to show that, $\forall \varepsilon > 0$, $\exists N$ such that

$$n > N \implies A_n \subset A^{(\varepsilon)}.$$

Suppose not, i.e., $\exists \varepsilon > 0$ and infinitely many A_{n_k} such that $A_{n_k} \setminus A^{(\varepsilon)} \neq \varnothing$. Let $A' := \bigcap_{k=1}^{\infty} (A_{n_k} \setminus A^{(\varepsilon)}) \subset (A^{(\varepsilon)})^c$, which is nonempty since $A_{n_k} \setminus A^{(\varepsilon)}$ is compact. On the other hand,

$$A' \subset \bigcap_{k=1}^{\infty} A_{n_k} = \bigcap_{n=1}^{\infty} A_n = A \subset A^{\varepsilon},$$

which leads to a contradiction.

(iii) Suppose $\{A_n\}$ is Cauchy in $(\mathrm{Cpt}(X), d_{\mathcal{H}})$. Let

$$B_n := \overline{\bigcup_{k=n}^{\infty} A_k}.$$

Claim: B_n is compact. Since (X, d) is complete, and B_n is closed, B_n is complete. By **Theorem 1.29.1**, it suffices to show that B_n is totally bounded. $\forall \varepsilon > 0$, $\exists N \geq n$ such that if $k \geq N$, then

$$d_{\mathcal{H}}(A_k, A_N) \leq \frac{\varepsilon}{4}.$$

In particular, $A_k \subset (A_N)^{(\varepsilon/4)}$. We can cover $\bigcup_{k=n}^{\infty} A_k$ by finitely many balls $B\big(x_k, \frac{\varepsilon}{2}\big), j = 1, \ldots, J$. Therefore

$$B_n = \overline{\bigcup_{k=n}^{\infty} A_k} \subset \bigcup_{j=1}^{J} B(x_j, \varepsilon),$$

i.e., B_n is totally bounded. Hence B_n is compact. As B_n is decreasing and compact, we can apply (ii) to show

$$B_n \xrightarrow{d_{\mathcal{H}}} \bigcap_{n=1}^{\infty} B_n := A.$$

We can obtain $A_n \xrightarrow{d_{\mathcal{H}}} A$ if we have

$$d_{\mathcal{H}}(A, B) \to 0.$$

As $A_n \subset B_n$, it suffices to show that $B_n \subset (A_n)^{(\varepsilon)}$ for all large n. Recalling that $\{A_m\}$ is a Cauchy sequence, $\forall \varepsilon > 0$, $\exists N$ such that

$$m, n \geq N \implies A_m \subset A_n^{(\varepsilon/2)}.$$

In particular, by **Lemma 1.29.1**,

$$B_n := \overline{\bigcup_{k=n}^{\infty} A_k} \subset \overline{A_n^{(\varepsilon/2)}} \subset A_n^{(\varepsilon)}.$$

(iv) Let A' be the set of limit points of convergence sequence $\{a_n\}$ with $a_n \in A_n$. It is easy to see that

$$B_n := \overline{\bigcup A_k} \supset A' \cup \left(\bigcup_{k=n}^{\infty} A_k \right),$$

which together with that $A = \bigcap_{n=1}^{\infty} B_n$ implies that $A' \subset A$. Now suppose $x \in A \setminus A'$. As $A_n \xrightarrow{d_{\mathcal{H}}} A$, $\forall k > 0$, $\exists N_k$ such that

(v) Let $\varepsilon > 0$. As X is totally bounded, we can find a finite set $X_\varepsilon \subset X$ such that X_ε is ε-dense in X.
Claim: The power set $\mathcal{P}(X_\varepsilon)$ is ε-dense in $(\mathrm{Cpt}(X), d_{\mathcal{H}})$. It suffices to show that, $\forall A \in \mathrm{Cpt}(X)$, $\exists E \in \mathcal{P}(X_\varepsilon)$ such that $d_{\mathcal{H}}(A, E) < \varepsilon$. We fix A, and define

$$E := X_\varepsilon \cap A^{(\varepsilon)},$$

then $E \subset A^{(\varepsilon)}$. Now for all $a \in A$, there exists $x_a \in X_\varepsilon$ such that $d(a, x_a) < \varepsilon$, therefore $x_a \in A^{(\varepsilon)}$. This means $x_a \in A^{(\varepsilon)} \cap X_\varepsilon = E$. In summary,

$$A \subset E^{(\varepsilon)} \implies d_{\mathcal{H}}(A, E) < \varepsilon.$$

(vi) Apply (iii), (v), and **Theorem 1.29.1**. □

1.30 Iterated function system

Let (X, d) be a metric space. $f : X \to X$ is a contraction if $\exists 0 \le \rho < 1$ such that

$$d(f(x), f(y)) \le \rho d(x, y) \quad \forall x, y \in X.$$

ρ is called **contraction constant**.

定理 1.30.1. *Let (X, d) be a complete metric space. Suppose $f : X \to X$ is a contraction with $\rho < 1$, then*

(i) $\exists! x \in X$ *such that* $f(x) = x$.

(ii) If $y \in X$, *we have*

$$d(x, f^k(y)) \le \rho^k d(x, y).$$

In particular, $f^k(y) \to x$ *as* $k \to \infty$.

Definition An **iterated function system**(IFS) on (X, d) is a finite family

$$\phi := \{\varphi_i\}_{i \in \Lambda}$$

of contractions. We say that ϕ has contraction constant ρ if each φ_i has contraction constant $\rho < 1$.

Example(Middle-α Cantor set) Let $\rho = \frac{1-\alpha}{2}$, and consider the IFS $\phi = \{\varphi_0, \varphi_1\}$, where $\varphi_0(x) = \rho x$, $\varphi_1(x) = \rho x + (1 - \rho)$. Alternatively, we can write for $i = 0, 1$,

$$\varphi_i(x) = \rho x + i(1 - \rho).$$

$I = [0, 1]$
$I_0 = [0, \rho] = \varphi_0(I), I_1 = [1 - \rho, 1] = \varphi_1(I)$
$i_1, i_2, \ldots, i_n \in \{0, 1\}$
$I_{i_1 i_2 \ldots i_n} = \varphi_{i_1} \circ \varphi_{i_2} \circ \cdots \circ \varphi_{i_n}(I).$
$C_{\alpha,n} := \bigcup_{i_1, \ldots, i_n} I_{i_1 \ldots i_n}$
$C_{\alpha,n+1} = \varphi_0(C_{\alpha,n}) \cup \varphi_1(C_{\alpha,n})$

$C_\alpha = \bigcap_n C_{\alpha,n}$

$\implies C_\alpha = \varphi_0(C_\alpha) \bigcup \varphi_1(C_\alpha)$.

$\phi(A) := \varphi_0(A) \cup \varphi_1(A)$

$\phi(C_\alpha) = C_\alpha$.

Fix $x \in C_\alpha$, and let $I^{(n)}(x)$ be the stage-n interval containing x, i.e.,

$$x \in I^{(n)}(x) = I_{i_1 \ldots i_n} = \varphi_{i_1 \ldots i_n}(I).$$

Then $I_{i_1 \ldots i_n} \to x$ as $n \to \infty$. Now

$$\begin{aligned}
\varphi_{i_1 i_2 \ldots i_n} &= \varphi_{i_1}(\varphi_{i_2 \ldots i_n}(y)) \\
&= \rho \varphi_{i_2 \ldots i_n}(y) + i_1(1-\rho) \\
&= \rho(\varphi_{i_3 \ldots i_n}(y) + i_2(1-\rho)) + i_1(1-\rho) \\
&= i_1(1-\rho) + i_2\rho(1-\rho) + \rho^2 \varphi_{i_3 \ldots i_n}(y) \\
&= \ldots \\
&= \sum_{k=1}^{n} i_k(1-\rho)\rho^{k-1} + \rho^n y
\end{aligned}$$

$$\implies x = \lim_{n\to\infty} \varphi_{i_1 \ldots i_n}(y) = \sum_{k=1}^{\infty} i_k(1-\rho)\rho^{k-1}.$$

$$C_\alpha = \left\{ (1-\rho) \sum_{k=1}^{\infty} i_k \rho^{k-1} : i_k \in \{0,1\} \right\}.$$

$\alpha = 0$: $\rho = \frac{1}{2}, C_\alpha = [0,1]$.

$\alpha = \frac{1}{3}$: $\rho = \frac{1}{3}$, $x \in C_\alpha$ if and only if x is of the form $\sum_{k=1}^{\infty} a_k 3^{-k}, a_k \in \{0,2\}$.

Let $\phi = \{\varphi_i\}_{i\in\Lambda}$ be an IFS with contraction constant ρ. Define

$$\tilde{\phi}(A) := \bigcup_{i\in\Lambda} \varphi_i(A) : \mathcal{P}(X) \to \mathcal{P}(X).$$

定理 1.30.2. *Let (X,d) be a complete metric space and ϕ an IFS with contraction constant $\rho < 1$. Then there exists a unique compact $K \subset X$ such that*

$$\tilde{\phi}(K) = K. \tag{1.44}$$

Moreover, for all compact $E \subset X$,

$$(\tilde{\phi})^n(E) \xrightarrow{d_{\mathcal{H}}} K$$

exponentially fast. If in addition $\tilde{\phi}(E) \subset E$, then

$$K = \bigcap_{n=1}^{\infty} (\tilde{\phi})^n(E).$$

证明. As φ_i is continuous and Λ is finite, $\tilde{\phi}$ is a map from $\mathrm{Cpt}(X)$ to $\mathrm{Cpt}(X)$. (1.44) means that K is a fixed point of $\tilde{\phi}$. By **Theorem 1.30.1**, it suffices to show that $\tilde{\phi}$ is a contraction. Let $\rho := \max_{i\in\Lambda} \rho_i < 1$. Suppose $d_{\mathcal{H}}(A, B) < \varepsilon$, then for all $i \in \Lambda$,

$$\varphi_i(A) \subset \varphi_i(B^{(\varepsilon)}) \subset [\varphi_i(B)]^{(\rho\varepsilon)},$$

which implies further that

$$\tilde{\phi}(A) \subset [\tilde{\phi}(B)]^{(f\varepsilon)}.$$

In summary,

$$d_{\mathcal{H}}(\tilde{\phi}(A), \tilde{\phi}(B)) \leq \rho\varepsilon,$$

i.e., $\tilde{\phi}$ is a contraction. It is easy to see that

$$d_{\mathcal{H}}(K, (\tilde{\phi})^n(E)) \leq \rho^n d_{\mathcal{H}}(K, E), \tag{1.45}$$

which yields the exponential convergence. Note that, by assumption, we have a decreasing sequence

$$E \supset \tilde{\phi}(E) \supset (\tilde{\phi})^2(E) \supset \cdots \supset (\tilde{\phi})^n(E) \supset \ldots,$$

therefore

$$(\tilde{\phi})^n(E) \xrightarrow{d_{\mathcal{H}}} \bigcap_{n=1}^{\infty} (\tilde{\phi})^n(E).$$

On the other hand, by (1.45), $(\tilde{\phi})^n(E) \to K$. Hence $K = \bigcap_{n=1}^{\infty} (\tilde{\phi})^n(E)$. □

Definition Let $\phi = \{\varphi_i\}_{i\in\Lambda}$ be an IFS. The compact set K satisfying

$$(\tilde{\phi})(K) = K$$

is called an **attractor** of ϕ.

Denote $i = i_1 \ldots i_n \in \Lambda^n$, and $\varphi_i = \varphi_{i_1} \circ \cdots \circ \varphi_{i_n}$. Given $i \in \Lambda^n$, we have $\varphi_{i_1\ldots i_n} K \subset \varphi_{i_1\ldots i_{n-1}} K$ as $\varphi_{i_n}(K) \subset K$. Using that $\varphi_{i_1\ldots i_n}$ has contraction constant ρ^n, we obtain that $\bigcap_{n=1}^{\infty} \varphi_{i_1\ldots i_n}(K)$ is non empty and consists of a single point x, which is denoted by $\phi(i)$, i.e., $\forall y \in K$,

$$\phi(i) = \lim_{n\to\infty} \varphi_{i_1\ldots i_n}(y). \tag{1.46}$$

It follows from the inequality $d(\varphi_{i_1\ldots i_n}(y), \varphi_{i_1\ldots i_n}(z)) \leq \rho^n d(y, z)$ that (1.46) holds for any $z \in X$. As $(\tilde{\phi})^n(K) \to K$, we see that $\phi : \Lambda^{\mathbb{N}} \to K$ is surjective.

命题 1.30.1. *Let $y \in X$, $\forall \varepsilon > 0$, $\exists N$ such that*

$$n \geq N \implies d(\varphi_i(y), K) < \varepsilon, \forall i \in \Lambda^n.$$

证明. Fix $x \in K$. Then $\varphi_i(X) \in K, \forall i \in \Lambda^n$. Choose N large enough such that

$$\rho^N \sup_{z\in K} d(y, z) < \varepsilon.$$

Then $\forall n \geq N, \forall i \in \Lambda^n$, we have

$$d(\varphi_i(y), \varphi_i(x)) \leq \rho^n d(y, x) \leq \rho^N \sup_{z\in K} d(y, z) < \varepsilon.$$

Since $\varphi_i(x) \in K$, this shows $d(\varphi_i(y), K) < \varepsilon$. □

Definition For $i, j \in \Lambda^{\mathbb{N}}$, let $N = N(i,j)$ be the largest number such that $i_1 \dots i_N = j_1 \dots j_N$. We define $d(i,j) = 2^{-N}$. View Λ as a discrete space, there exists a natural product topology τ on $\Lambda^{\mathbb{N}}$.

命题 1.30.2. *Let τ_d be the topology induced by d. Then $\tau_d = \tau$.*

命题 1.30.3. *Fix $i, j \in \Lambda^{\mathbb{N}}$, and denote $N = N(i,j)$. Then*

$$d(\phi(i), \phi(j)) \le \rho^N \operatorname{diam}(K).$$

In particular, ϕ is Hölder continuous.

证明. Fix $x \in K$. $\forall n \ge N$, we have

$$\begin{aligned} & d(\varphi_{i_1 \dots i_n}(x), \varphi_{j_1 \dots j_n}(x)) \\ = & d(\varphi_{i_1 \dots i_N}(\varphi_{i_{N+1} \dots i_n}(x)), \varphi_{i_1 \dots i_N}(\varphi_{j_{N+1} \dots j_n}(x))) \\ \le & \rho^N d(\varphi_{i_{N+1} \dots i_n}(x), \varphi_{j_{N+1} \dots j_n}(x)) \\ \le & \rho^N \operatorname{diam}(K). \end{aligned}$$

□

Given $i = i_1 \dots i_k \in \Lambda^k$, we define the cylinder set $[i] \subset \Lambda^{\mathbb{N}}$ by

$$[i] := \{j \in \Lambda^{\mathbb{N}} : j_1 \dots j_k = i_1 \dots i_k\}.$$

引理 1.30.1. *For all $[i]$ with $i \in \Lambda^k$, we have*

(i) *$[i]$ is both open and closed in $\Lambda^{\mathbb{N}}$;*

(ii) $\operatorname{diam}([i]) = 2^{-k}$;

(iii) $[i] = B_{2^{-k}}(j), \forall j \in [i]$;

(iv) *The family of all cylinder sets forms a basis for the topology on $\Lambda^{\mathbb{N}}$.*

证明. (i) Since (by the definition of product topology) $\pi_k : \Lambda^{\mathbb{N}} \to \Lambda^k$ is continuous and $[i] = \pi_k^{-1}(i)$, $[i]$ is both open and closed;

(ii) $\forall j, j' \in [i]$, we have

$$d(j, j') \le 2^{-k},$$

and

$$d(ii_{k+1}, ij_{k+1}) = 2^{-k}$$

when $i_{k+1} \ne j_{k+1}$;

(iii) The proof is contained in (ii);

(iv) omit. □

引理 1.30.2. *$\forall j \in \Lambda$, define a map $\tilde{\varphi}_j : \Lambda^N \to \Lambda^{\mathbb{N}}$ by*

$$\tilde{\varphi}_j(i_1 i_2 \cdots) = j i_1 i_2 \cdots .$$

Then, $\forall j \in \Lambda$ and $\forall i \in \Lambda^{\mathbb{N}}$, we have

$$\phi(\tilde{\varphi}_j(i)) = \varphi_j(\phi(i)) \in X.$$

证明. We recall that $\phi(i) = x$ means that

$$\lim_{n\to\infty} \varphi_{i_1\dots i_n}(y) = x \in K.$$

Therefore, since φ_j is continuous,

$$\begin{aligned}\varphi_j(\phi(i)) &= \varphi_j(\lim_{n\to\infty} \varphi_{i_1\dots i_n}(y)) \\ &= \lim_{n\to\infty} \varphi_{ji_1\dots i_n}(y) \\ &= \phi(\tilde{\varphi}_j(i)).\end{aligned}$$

□

Recall that

$$\operatorname{supp}\mu := X \setminus \bigcup U, U \in \{U : U \text{ open}, \mu(U) = 0\}.$$

Definition A vector $p = (p_i)_{i\in\Lambda}$ is called a **probability vector** if $p_i \geq 0, \forall i \in \Lambda$, and $\sum_{i\in\Lambda} p_i = 1$. We say p is **positive** if $p_i > 0$ for all $i \in \Lambda$.

定理 1.30.3. *Let $p = (p_i)_{i\in\Lambda}$ be a probability vector. Then there exists a unique Borel probability measure $\mu := \mu_{\phi,p}$ on K such that*

$$\mu = \sum_{i\in\Lambda} p_i \cdot \varphi_i\mu,$$

where $\varphi_i\mu = (\varphi_i)_\sharp$. If p is positive, then $\operatorname{supp}\mu = K$.

证明. Let p be a Borel probability measure on Λ with $p(\{i\}) = p_i, \forall i \in \Lambda$. Then there exists a unique Borel measure $\tilde{\mu}$ on $\Lambda^{\mathbb{N}}$ such that for all $n_1, \dots, n_k \in \mathbb{N}$ and $E \subset \Lambda^k$, we have

$$\tilde{\mu}(\pi^{-1}_{(n_1,\dots,n_k)}(E)) = \underbrace{(p \times \cdots \times p)}_{k \text{ times}}(E),$$

where $\pi_{(n_1,\dots,n_k)}(j) = j_{n_1} \dots j_{n_k}, \forall j \in \Lambda^{\mathbb{N}}$. As $E = \bigcup_{i\in\Lambda} \tilde{\varphi}_i\tilde{\varphi}_i^{-1}(E)$, we have

$$\tilde{\mu}(E) = \sum_{i\in\Lambda} p_i \cdot \tilde{\mu}(\tilde{\varphi}_i^{-1}(E)) \tag{1.47}$$

for all cylinder sets E. Since the σ-algebra of $\Lambda^{\mathbb{N}}$ is generated by cylinder sets, (1.47) holds for all Borel sets $E \subset \Lambda^{\mathbb{N}}$. Recalling that $\phi : \Lambda^{\mathbb{N}} \to K$, we denote

$$\mu := \phi\tilde{\mu} = (\phi)_\sharp\tilde{\mu}$$

Apply ϕ to (1.47) to obtain

$$\tilde{\mu}(\phi^{-1}(F)) = \sum_{i\in\Lambda} p_i\tilde{\mu}(\tilde{\varphi}_i^{-1}\phi^{-1}(F)),$$

i.e.,

$$\mu = \sum_{i\in\Lambda} p_i\phi\tilde{\varphi}_i\tilde{\mu} \overset{\text{Lemma 2.7}}{=\!=} \sum_{i\in\Lambda} p_i\varphi_i\phi\tilde{\mu} = \sum_{i\in\Lambda} p_i\varphi_i\mu.$$

□

1.31　讲义: 几何测度论/ 习题

原网页

1.32　Dimensions

1. Let $\mathcal{A} \subset 2^X$ and $\mu : \mathcal{A} \to [0, \infty]$ be such that $\emptyset \in \mathcal{A}$, $X \in \mathcal{A}$ and $\mu(\emptyset) = 0$. For any set $A \subset X$, define
$$\mu^*(A) = \inf \left\{ \sum_{j=1}^{\infty} \mu(E_j) : E_j \in \mathcal{A} \text{ and } A \subset \bigcup_{j=1}^{\infty} E_j \right\}.$$
Show that μ^* is an outer measure.
2. Show that if μ is a measure, $A \subset B$, B is μ-measurable and $\mu(A) = \mu(B) < \infty$, then $\mu(A \cap C) = \mu(B \cap C)$ for all μ-measurable sets C.
3. A nonempty collection of subsets $P \subseteq 2^X$ is a π-system provided $A, B \in P$ implies
$$A \cap B \in P.$$
A collection of subsets $\mathcal{L} \subseteq 2^X$ is a λ-system provided
   (i) $X \in \mathcal{L}$;
   (ii) $A, B \in \mathcal{L}$ and $B \subseteq A$ implies $A \backslash B \in \mathcal{L}$
   (iii) if $A_k \in \mathcal{L}$ and $A_k \subseteq A_{k+1}$ for $k = 1, \ldots,$ then
$$\bigcup_{k=1}^{\infty} A_k \in \mathcal{L}.$$
Prove the following results:
   (i) If P is a π-system and $\mathcal{L}$ is a λ-system with $P \subseteq \mathcal{L}$, then
$$\sigma(P) \subseteq \mathcal{L}.$$
   (ii) Let μ and ν be two finite Borel measures on $\mathbb{R}^n$ such that
$$\mu(R) = \nu(R)$$
for all closed "rectangles"
$$R := \{x \in \mathbb{R}^n | -\infty \leq a_i \leq x_i \leq b_i \leq \infty (i = 1, \ldots, n)\}$$
Then
$$\mu(B) = \nu(B)$$
for all Borel sets $B \subseteq \mathbb{R}^n$.

1.33 Orthogonal Projections

1. Let X be a locally compact Hausdorff space, K a compact subset of X, and $\{U_i\}_{i=1}^n$ an open covering of K. Show that there is a partition of unity on K subordinate to $\{U_i\}_{i=1}^n$ consisting of compactly supported functions.
2. Let μ be an α-Frostman measure on $\mathbb{R}^n$ with $\alpha > n-1$, and $B(0,R) := \{x \in \mathbb{R}^n : |x| \le R\}$. Show that
$$\mu(\partial B(0,R)) = 0.$$
3. Suppose μ is an α-Frostman measure. Let $\psi \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ satisfy $\psi \ge 0$ and $\int \psi = 1$. Show that $\mu_\varepsilon := \psi_\varepsilon * \mu$ are α-Frostman measures (uniformly in ε).
4. Suppose that μ is an α-Frostman measure on E. Show that there exist two compact subsets K_1 and K_2 with $\operatorname{dist}(K_1, K_2) > 0$ such that $\mu|_{K_1}$ and $\mu|_{K_2}$ are α-Frostman measures.
5. Let μ and ν be finite Radon measures on $\mathbb{R}^n$. Show that $\mu * \nu$ is a Radon measure.

1.34 Differentiation of measures

1. Let $\varepsilon > 0$ and $N \in \mathbb{N}$. Show that there exist disjoint balls $\{B_j\}$ in $\mathbb{R}$ such that
$$\sum_j \chi_{(1+\varepsilon)B_j} \ge N$$
on a set of positive Lebesgue measure.
2. Let μ and λ be Radon measures on $\mathbb{R}^n$. Show that $D(\mu, \lambda, x)$ is a Borel function.
3. Let μ and λ be Radon measures on $\mathbb{R}^n$. Suppose that $\mu \ll \lambda$. Show that
$$\int D(\mu,\lambda,x)^2 \, d\lambda x = \int D(\mu,\lambda,x)\, d\mu x.$$
4. Let μ and λ be Radon measures on $\mathbb{R}^n$. Find μ and λ such that
$$\int_B D(\mu,\lambda,x)\, d\lambda x < \mu(B)$$
for some Borel set B.

1.35 Fourier Transforms

1. **(du Bois Reymond lemma)** Suppose that $f \in L^1_{loc}(\mathbb{R}^n)$ and
$$\int f\varphi = 0 \quad \forall \varphi \in C_c^\infty(\mathbb{R}^n). \tag{1.48}$$
Then $f = 0$ a.e. In particular, if f is further assumed to be continuous, then $f = 0$.
2. Let $f(x) = e^{-\pi|x|^2}$ with $x \in \mathbb{R}^n$. Show that $\widehat{f}(\xi) = f(\xi)$.
3. Let $\mu \in M(\mathbb{R}^n)$. If there exists a constant C such that, for some $s > 0$, we have
$$\mu(B(x,r)) \ge Cr^s \quad \forall x \in \operatorname{supp}\mu, \quad 0 < r < 1.$$
Show that, for $R > 1$,
$$\int_{B(0,R)} |\widehat{\mu}(\xi)|^2 d\xi \ge CR^{n-s}.$$

4. **[Steinhaus' s Theorem]** Let E be a measurable subset of $\mathbb{R}$ such that its Lebesgue measure is positive. Then
$$\Delta'(E) := \{x - y : x, y \in E\}$$
contains an open interval containing 0.
5. Suppose $\mu, \mu_\varepsilon, \varepsilon > 0$ are Radon measures on $\mathbb{R}^n$, and $\mu_\varepsilon \rightharpoonup \mu$. Show the following results.
   (i) Let K be a compact subset of $\mathbb{R}^n$. Then $\sup_{\varepsilon>0} \mu_\varepsilon(K) < \infty$.
   (ii) $\mu_\varepsilon \times \mu_\varepsilon \rightharpoonup \mu \times \mu$.
6. Let $\mathcal{H}_k$ be the Hankel transform
$$\mathcal{H}_k(F)(t) = \int_0^\infty J_k(tr) f(r) r \, \mathrm{d}r.$$
Show that
$$\mathcal{H}_k \mathcal{H}_k f = f.$$

1.36　Self-similar sets

1. Let $0 < \varepsilon < \delta$. Show that $\overline{A^{(\varepsilon)}} \subset A^{(\delta)}$, where $A^{(\varepsilon)} := \{x \in X : d(x, A) < \varepsilon\}$.
2. Let $\{\ell_j\}$ be a sequence of lines in $\mathbb{R}^2$ such that
   (i) $\ell_j \cap B(0,1) \neq \emptyset$ for any j,
and (ii) $\ell'_j := \ell_j \cap B(0,2)$ is convergent under the Hausdorff metric.
   Show that the limit of ℓ'_j is of the form $\ell \cap B(0,2)$ for some line ℓ.
3. Let Λ be a finite set. For $i, j \in \Lambda^{\mathbb{N}}$, we define $d(i,j) = 2^{-N}$, where N is the largest number such that
$$i_1 \cdots i_N = j_1 \cdots j_N.$$
Let τ_d be the topology on $\Lambda^{\mathbb{N}}$ induced by d, and τ the standard product topology on $\Lambda^{\mathbb{N}}$. Show that $\tau = \tau_d$.

1.37　试卷: 几何测度论

原网页

2025 秋

1. (i) 请陈述 Frostman 引理。
   (ii) 设 $\alpha > n-1$, μ 是 $\mathbb{R}^n$ 上的 α-Frostman 测度。$\mathbb{S}^{n-1} = \{x \in \mathbb{R}^n : |x| = 1\}$. 证明：
$$\mu(\mathbb{S}^{n-1}) = 0.$$
2. (i) 请陈述 Minkowski 维数的定义。
   (ii) 设 k 是正整数，$S_k := \{k^{-n} : n \in \mathbb{Z}_{\geq 0}\}$. 计算 S_k 的 Minkowski 维数。
3. 设 μ, λ 是 $\mathbb{R}^n$ 上的 Radon 测度，且 $\mu \ll \lambda$. 证明：
$$\int D(\mu, \lambda, x)^2 \, \mathrm{d}\lambda(x) = \int D(\mu, \lambda, x)^2 \, \mathrm{d}\mu(x).$$

4. (i) 设 μ 是一个 s-Frostman 测度，$0 < s < n$. 证明：

$$\int_{B(0,R)} |\hat{\mu}(x)|^2 \,\mathrm{d}x \leq CR^{n-s}.$$

(ii) 设 μ 是一个紧支撑有限 Radon 测度，满足对任意的 $R \geq 1$, 成立

$$\int_{B(0,R)} |\hat{\mu}(x)|^2 \,\mathrm{d}x \leq CR^{n-s},$$

证明：对任意的 $t \in (0, s)$, 有 $I_t(\mu) < \infty$.

5. (i) 请陈述迭代函数系统的强分离条件和开集条件的定义。
(ii) 证明：强分离条件蕴含开集条件。

2024 春

1. (i) 请陈述 Hausdorff 测度的定义。
(ii) 设 $0 < r < \infty$。证明 $\mathcal{H}(B(x,r)) \subset (0,\infty)$，由此证明存在 $c \in (0,\infty)$ 使得 $\mathcal{H}^n \subset c\mathscr{L}^n$。
2. (i) 设 μ 是 α-Frostman 测度。设$\Psi \in C_c^\infty(\mathbb{R}^n)$ 满足$\Psi \geq 0$ 且 $\int \Psi = 1$。证明 $\mu_\varepsilon := \Psi_\varepsilon * \mu$ 都是 α-Frostman 测度（一致）。
(i) 设 μ 是 E 上的 α-Frostman 测度。证明存在 E 上的两个紧集 K_1 和 K_2，使得 $\mathrm{dist}(K_1, K_2) > 0$，且 $\mu|_{K_1}$ 和 $\mu|_{K_2}$ 是 α-Frostman 测度。
3. 设 λ, μ 是 $\mathbb{R}^n$ 上的 Radon 测度。
(i) 设 $\mu \ll \lambda$。证明

$$\int_B D(\mu, \lambda, x) d\lambda = \mu(B).$$

(ii) 找 μ, λ，使得对某个 B 有

$$\int_B D(\mu, \lambda, x) d\lambda < \mu(B).$$

4. 称一个测度是**连续的**，是指每个点都是零测的。
设 $\{\mu_N\}, N = 1, 2, \ldots$ 和 ν 是 $\mathbb{R}$ 上的非负 Radon 测度，ν 是连续的。若 $\widehat{\mu}_N(\xi) \to \widehat{\nu}(\xi)$ 当 $N \to \infty$，对所有的 $\xi \in \mathbb{R}$，证明 $\mu_N((a,b)) \to \nu((a,b))$ 对所有的 $a < b$。
5. (i) 请陈述迭代函数系（IFS）吸引子 K 的定义，并验证其是良定义的。
(ii) 对于概率测度 $p = p_i, i \in \Lambda$，证明在吸引子 K 上存在 Borel 概率测度 μ 满足

$$\mu = \sum_{i \in \Lambda} p_i (\varphi_i)_\# \mu.$$